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Transition to Advanced Mathematics

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Preface

In the spring of 2011, I had a revelation. I was teaching both abstract algebra and its prerequisite discrete math. All of my students in abstract algebra had supposedly completed the course that would permit them to work as somewhat independent proof-writers. Yet, they seemed to struggle when not provided with a specific template to copy. At the same time, I experienced what I now informally call the "Euler-ϕ" phenomenon in discrete math. I had assigned a homework problem in which the Euler-ϕ function was defined and accompanied with some very straightforward questions. More than half the class skipped the problem entirely because they assumed I had made a mistake in assigning it, since we had not already covered the Euler-ϕ function in class. The belief that they couldn't possibly be asked to face a question they hadn't already been shown how to solve felt unacceptable to me. I finished the semester realizing something absolutely needed to change. The timing was excellent, as that summer I headed to MathFest and ended up sitting next to Ed Burger on the plane. As a fairly new and inexperienced faculty member, I innocently asked him if he'd had experience teaching with IBL. In fact, I could hardly have been sitting next to someone with more experience! That conversation got the wheels turning. I then had the luck to be introduced to Ron Taylor at that MathFest, another experienced IBL user, and he generously allowed me to pick his brain about IBL and gave me a nudge as an unofficial mentor. Yet another helpful discussion came from Patrick Rault, who himself had recently taken the plunge and was able to assure me that it would be okay. Armed with all this motivation, I decided I was just going to dive into the deep end, write my own IBL notes, and give it a try. The first semester ended up being even more successful than I'd hoped, and once I went down this road I knew I was never going back. These notes have undergone many revisions since that time (along with my pedagogical choices), but what you are about to read are essentially the fruits of that initial labor. I hope you will find them useful!

To the Instructor

Welcome to my version of the course aimed at transitioning our undergraduate math majors to advanced mathematical concepts and argumentation. The notes you will read below consist largely of standard IBL sequencing designations: Definition, Exercise, Problem, Theorem, Remark, Example. “Exercises” are typically designed to introduce and reinforce meta-skills such as interpreting and processing a new definition, whereas “Problems” are are generally aimed at full utilization and application of a definition/concept. Notes to the instructor appear throughout the text, designated as superscripts and detailed at the end.

You may notice that there is more exposition than in a typical IBL text, but that choice was made within the following framework:

- I never lecture, literally never. This is a deliberate choice. I have found in the past that when I lectured even a tiny bit in this course, any wording I provided would inevitably find its way into someone’s homework. The belief that the words I put on the board are gospel, to be copied and reproduced at every opportunity, seems to require a zero-tolerance policy and multiple semesters to be broken. I have also found that students have sometimes concluded that whatever understanding they had developed of a concept until that point was “wrong” because it did not entirely match my way of expressing it, whether or not that was actually the case.
- The amount of exposition decreases as the text moves forward. For example, proofs are introduced in chapter 2, where there are several pages of remarks and examples. However, that level of exposition regarding proofs is deliberately avoided for the most part in the chapters that follow, with an exception for induction.

0.1 About the Course and the University

The students for whom these notes were written represent a wide range of abilities and preparation. I have successfully used this text for anywhere from graduate-school bound students to non-STEM majors who have barely scraped through their previous math courses. There are generally several uniting factors among them:

they want to work hard, have many commitments outside of school including work and families that might prevent them from doing so, and know they have a lot to learn. They are friendly, joyful, and open-minded, which makes them a pleasure to work with despite any mathematical shortcomings with which they may enter the course.

While the name of this course at my institution is specifically “Discrete Mathematics”, we actually mean for it to serve more as an introduction-to-advanced-mathematics type of course. In fact, we also have a Discrete Math for Computer Science course which is more computational and does not count for credit for the mathematics major. We kept the name “Discrete Math” for accreditation reasons for computer science majors, so that double majors in both math and computer science could use this course for both, but it is important to keep in mind as you read through the notes that this course is mainly intended for math majors, and is most definitely meant to serve as our “transitions” course. That was very much in my mind in writing these notes, that the main goal was to prepare students for further study in mathematics. In that sense, there are some places in which I have been more prescriptive (such as in the introduction to proofs in Chapter 2), but I believe I have still left plenty of room to “play”.

This course has only a Calculus I prerequisite, which is mainly for maturity purposes, and is meant to be taken fairly early within the student’s career. You will notice that there is only a smattering of calculus content (and even pre-calculus content) within the notes, and those problems could be easily removed or modified to eliminate the prerequisite. This course is itself a prerequisite for most of our upper-level courses, including linear algebra, abstract algebra, and real analysis. The class size typically ranges from 10 to 20 students. CCSU is a mid-sized public university with students of very mixed backgrounds, including many transfer students and/or students who completed all previous math classes at community college. We therefore often have very little control over their preparedness, and we have come to expect a very wide range in background in any given class. This requires extra fine-tuning in terms of grading and moderating during presentations.

0.2 Class Background and Structure

I have used these notes for classes in both two-day-a-week and three-day-a-week structures, both on-ground and online synchronously. Thus, in the descriptions that follow of use of class time, I avoid specifying a certain number of minutes in any particular setting, as it will adjust with the length of the course. My personal preference is for longer class periods that meet fewer times, but I have not noticed a significant difference in outcomes in either case. I will also focus mainly on the on-ground structure vs. online, as it is fairly easily adaptable to online synchronous if there is access to breakout groups. Each class period is mainly split into two parts: presentations/discussion of presentations, which typically takes about a quarter to a third of the period, followed by group work for the remainder of the time.

0.3 Expectations and Grading

I will frame this section by first discussing my overarching expectations for students who successfully complete this course. I expect that they complete the course being able and willing to do the following (ideally by default):

- Jump in and start the problem-solving process, without waiting to be told what to do.
- Assess their own reasoning and understanding to the best of their ability, and assess the reasoning of peers. At least not be afraid to try.
- Make valid and complete logical arguments based on their own conceptual understanding, which may or may not yet be elegant.

These goals underly every interaction I have with students, and every comment I provide (or don't provide) on written and oral work. I am willing to sacrifice content coverage and perfectly-written solutions in order to allow students room to grow on their own as problem-solvers and communicators.

0.3.1 Presentations

My expectation of presenters is that they are prepared to present what they believe is a correct solution. The grading is only on a P/NP (pass/not pass) basis, and very few presentations receive an "NP". (In those cases, students are allowed to post a follow-up presentation online to boost the grade to a P - see more details in Section 0.4 regarding online presentations.) Passing is based on their demonstrated correctness and mathematical understanding, a large component of which is their response to audience questions. This serves the dual purpose of rewarding those who have truly understood their problem, even if they made some errors, while not rewarding those who might have copied the solution from another source or followed along during group work without understanding. Live presenters are allowed to correct their mistakes on the spot with essentially no penalty. Clarity of presentation is also taken into account, not to be confused with elegance, the key goal in this category being clear communication of their argument to their peers. Typical public-speaking criteria such as board work and projection are not officially part of the grade, though often times classmates will offer (gentle) suggestions regarding these aspects.

Being an attentive and responsive audience member is also part of the evaluation criteria. All questions and constructive contributions are encouraged, even the low-hanging fruit such as typos or spelling errors, as they are generally done in good faith and with respect according to the atmosphere we create. Comments that suggest a different approach to solving the problem are generally not permitted except when explicitly solicited. Lastly, note-taking is not allowed.

Students' job as online audience members is to provide regular feedback on posted presentations and uphold relevant comment threads. As there can be a tendency to offer simplistic comments along the lines of "Good job! Clear slides," in this setting, I emphasize that only "substantive feedback" counts officially toward participation. Accordingly, I provide the following description of what qualifies as such:

- Correction of an error
- A request to clarify
- An observation that ties the problem to a different mathematical concept in the course
- An exploratory question such as "I wonder what would happen if. . . ."
- Sharing of a different approach (but only when solicited)

Other comments such as "Good job!" or "How did you do that in LaTeX?" are certainly welcome. They just don't count as "substantive feedback".

0.3.2 Written Homework

Written homework is meant to be submitted regularly and have both a summative and formative role via rewrites. No specific problems are assigned, but certain thresholds must be met by the end of the semester for each grade category (more details below). Note that only "Problems", and not "Exercises", are submitted. Every student is officially required to submit at least three new problems per week for at least 10 weeks of the semester to pass, though behind the scenes I have waived that requirement for those that satisfactorily demonstrated learning without having met it. There is also a maximum of six problems per week (including rewrites). Within this framework, students are free to choose which problems they submit.

Each problem is assigned an "M", "P", or "N", via the description below taken directly from my syllabus:

M – Mastered: you've got the concept and you're good to go (worth 2 points)
P – Progressing: there's still something important that you're missing but you're on your way (worth 1 point)
N – Not yet: you're missing the overall concept and should go back and maybe start fresh in your understanding of the concept (worth 0 points)

Problems may be rewritten as many times as a student desires in order to achieve an “M”; however, after a given problem has been presented, it can achieve at most a “P*” (worth 1.1 points) if completed at an “M” level.

The numerical values are not written on the page as the grade presented to students, but are used only within my gradebook for easier bookkeeping. The final required totals for each category are detailed at the end of this section.

0.3.3 Portfolios

Students have a choice between taking a final exam or submitting a portfolio. A complete final portfolio consists of ten “chapters”, each chapter consisting of a problem completed at an M or P* level, all supporting work and rewrites including photos of group work, and a discussion of their problem-solving and learning process from that problem. I am careful to distinguish this from simply listing mistakes and their corrections - the “why” is required. Each chapter is graded as “T” for “thorough”, “S” for “satisfactory”, or “I” for “incomplete”, which are then converted into 2, 1, and 0 points respectively.

0.3.4 Overall Assessment for Course Grades

Given the assessment categories described above, below are the specifications for each grade category, taken directly from my syllabus:

To earn an A you must do all of the following:

- Earn at least 95 points in the “Homework” category
- Earn at least 13 homework points in each chapter
- Present at least six times with a grade of P, in at least four different chapters
- Earn a grade of at least 17/20 on the portfolio, or at least an 85% on the final exam
- Provide substantive feedback on at least one-third of the presentations (according to the “Substantive Feedback Guidelines”)

To earn a B you must do all of the following:

- Earn at least 80 points in the “Homework” category
- Earn at least 11 homework points in each chapter
- Present at least four times with a grade of P, in at least three different chapters
- Earn a grade of at least 15/20 on the portfolio, or at least an 75% on the final exam

- Provide substantive feedback on at least one-quarter of the presentations (according to the "Substantive Feedback Guidelines")

To earn a C you must do all of the following:

- Earn at least 65 points in the "Homework" category
- Earn at least 9 homework points in each chapter
- Present at least two times with a grade of P, in at least two different chapters
- Earn a grade of at least 13/20 on the portfolio, or at least an 65% on the final exam
- Provide substantive feedback on at least one-sixth of the presentations (according to the "Substantive Feedback Guidelines")

To earn a D you must do all of the following:

- Earn at least 50 points in the "Homework" category
- Earn at least 7 homework points in each chapter
- Present at least once with a grade of P
- Earn a grade of at least 10/20 on the portfolio, or at least an 55% on the final exam
- Provide substantive feedback on at least several presentations (according to the "Substantive Feedback Guidelines")

If you do not meet all of the D requirements, you will receive an F.

The distinctions of "+". within each letter grade will be based on "extra" achievements such as more presentations and/or homework problems, as well as your participation in class. If you have achieved all but one level in a given category (and that level is still at the requirement for the grade level immediately below it), you will receive the "-" grade for that letter.

I reserve the right to make allowances in determining an individual's final grade at the end of the semester based on the understanding I have seen demonstrated; however, that should not be counted on.

You might have noticed that the standards to receive a "D" are fairly low, as students must earn at least a "C-" in order to receive credit for the course within the major at our institution.

As a final note, in practice I am not necessarily that thorough about counting the number of times a student has provided substantive feedback. It is usually easy enough to determine where a student falls according to the other criteria and then make the final call based on my overall impression of their participation level. In general, if a student has performed strongly in all categories except this one, I will let it slide. I also reserve the right to adjust on a case-by-case basis (always up, never down) if a student has demonstrated learning to me despite not hitting every required level.

0.4 Logistics of Presentations

In my use of this text, almost all of the problems in the notes are presented, while exercises are not presented. The problems that I choose not to be presented can vary from semester to semester. I typically aim for six presentations per week, or two-to-three during/between class periods. These are split roughly 50-50 between in-class presentations in real time and video presentations posted on the online platform GoReact, where classmates respond with timestamped comments. Most, if not all of the discussion of the presentations posted on GoReact is completed there, so that for those I only take class time to wrap up loose ends or highlight a particularly important point. This allows us time to dive deeply into the presentation(s) done in real time. Here I encourage detailed comments, all the way down to spelling and word choice. Students sign up for presentations in advance via a schedule in a google spreadsheet, so they are usually reasonably well prepared. For the in-person presentations, I use a receipt book with carbon copy to jot down comments so that they can receive feedback immediately following.

0.5 Group Work

I believe the group work to be an essential part of the pedagogy, which is why I make sure to dedicate ample time to it each class period. While there are many students who would be able to read through these notes independently and successfully master the concepts within, most require the opportunity to discuss, correct, and fine-tune their understanding with peers. In this sense, I see the group work as what levels the playing field between those who can easily learn by reading and those who struggle more with that. That being said, in my experience students who rely *only* on group work for their learning will not be successful.

As previously noted, the group work portion always takes place immediately after the presentations. It is entirely unstructured, meaning the students break into groups and choose on their own what they would like to work on. (I do however encourage them at the end of class to decide as a group the point in the notes where they will aim to have prepared for the next class.) I allow the students to choose their own groups, and in general let them keep those groups throughout the semester unless I observe or am made aware of a dynamic that needs to be changed.

I also encourage students to move around the room and work with other classmates at any time during group work, in particular if their group is working on a different problem than they are. I purposely select a classroom with boards on several walls so that groups may each have their own board space if desired (which sometimes takes encouragement on my part). Typically, if students wander around it is to either compare the work they have already done with others', or to seek a kick start if they're stuck, but they generally tend to stay in their own groups the vast majority of the time.

0.6 Proof Grading Codes

The idea for this stemmed from Anders Hendrickson's excellent *Elements of Style for Proofs*[1]. However, I didn't have the energy to create a document so detailed or to adapt his to my approach. I did, however, appreciate the time-saving aspect of being able to name common errors by number, or "code". I therefore created a quick, short list of what I find to be the most common errors among my students. It saves quite a bit of time! It also has the added advantage of highlighting to students when they are making the same mistake repeatedly, and according those mistakes the appropriate level of importance (important enough to be acknowledged with their own code, but not as important as ones that receive full comments).

AN: Pronoun without a clear antecedent (as in "it" or "their" but it's not clear what they refer to)
AP: Assuming what you want to prove
AU: Already used symbol
BA: Argument is written backward
CS: Use complete sentences
IN: Incorrect notation
IT: Incorrect terminology
LC: Need a logical connector word or words (have a mathematical fact without a logical lead-in)
NC: Not clear what you mean (can't follow)
NW: Too many symbols (need words mixed in)
NP: Writing that something is true before you've shown it (not yet proven)
PE: "Proof" by example (proving a "for all" using just one example)
SC: Need a specific counterexample
SP: State the problem
UC: Unjustified claim or conclusion
UN: Undefined or unintroduced symbol, object, or claim
VS: Vacuous statement (e.g. $1 = 1$)
WO: Write the symbol out as a word
WC: Word choice is logically incorrect

[1] Book chapter, *Resources and Pedagogical Techniques for Enhancing the Teaching of Proof-Writing Across the Curriculum*, MAA Notes Volume

0.7 LaTeX and Piazza

I require all written homework to be submitted in LaTeX (via the free templates provided on the website Overleaf). In my opinion, this course is the ideal point to start learning it since there are so few symbols required in the beginning. I used to use one class period in the second week of the semester to take students to the computer lab to get them started, but when we were forced online I had to have them learn from a training video I made instead. They learned so well that I ended up just using that once we returned to campus; thus far they have continued to pick it up very quickly from the video. After that, they essentially use google (quite skillfully) to progress from there. In my experience students find it a reasonable request and are happy to have learned the skill. It also makes it quicker and easier for them to do rewrites.

They might also ask questions regarding LaTeX or any other aspect of the course on Piazza, which is what I use as our main online course platform. It is free for small classes, simple to use, and accepts LaTeX code. It is also linkable from Blackboard. I use it to post announcements, course documents, and to answer questions. I also use it to collect, grade, and return homework submissions. I currently have no official structure built into the course that requires students to post regularly, but it would be easy to implement such a system using Piazza.

To the Student

Good mathematics is not about how many answers you know, but about how you behave when you don't know.

Welcome to upper-level mathematics and advanced mathematical thinking! In this course you will learn not just new mathematical content, but how to problem-solve and make sense of new concepts for yourself.

These notes are meant to be self-guided, and include Definitions, Theorems, Remarks, Exercises, and Problems. They are not very long, considering they represent an entire course. Therefore, it is expected that you, the reader will read everything, and not skip to the problems as a student might often do with a long text. It is in fact imperative that you carefully read every part of the text, in order, and complete the exercises before attempting the problems. If you skip right to the problems, you will find that you are unprepared and will struggle to solve them.

Note that most of the time you will be asked to solve problems for which you have been given no similar example, but which follow straight from given definitions and exercises. In this subject (and in higher math in general), each problem is different, and thus there is no "template"; therefore, an important part of your learning is the skill of unpacking and applying mathematical statements and concepts for your own use. An analogy you might keep in mind is that you are learning to cook, not follow recipes, or that you are learning to be furniture-makers, not chair- or table-makers only. With this in mind, when approaching each problem, it won't be fruitful to search through the notes for a similar example. Instead, the first questions you should always ask yourself are, "Do I understand what all the words/symbols/terms in the problem mean? Have I done the exercises using these terms?" If not, your first step should be to revisit the text itself, until you fully understand every term/concept and have already practiced using them on the exercises.

It may seem intimidating at first to imagine mastering an entire course's worth of material without being *told*. But, you will be surprised at how much you can accomplish and learn when you are free to think for yourself. This course is heavily based on logic, which you naturally possess as a human being, and so you will be able to tap into what you already understand from simply being a citizen of our world. In addition, you will gain a great deal of knowledge about *how* you learn, not simply the *facts* that you learned. You will be able to carry that skill not only to subsequent courses but throughout the rest of your life, long after you have left school and would have a professor by your side feeding you the correct answer. I firmly believe by the end of the course you will be proud of all you have accomplished, and will be ready to move forward into the mathematical world and beyond.

Acknowledgements

I must first immediately thank those that inspired me to take the plunge in the first place into an IBL approach: Ron Taylor, Ed Burger, and Patrick Rault (though Ed Burger is probably unaware of his influence on me). I would like to thank Ron Taylor in particular for allowing me to pick his brain and steal his ideas, and Patrick Rault for reassuring me that my students wouldn't hate me.

I am grateful to Ileana Vasu and Moshe Cohen for fruitful discussions regarding the subject and for (unwittingly) contributing problems that found their way into the text. I would also like to acknowledge the instructors at the Université Paris Diderot for their department exercise sheets, which contained several interesting problems I have chosen to adapt and include here (indicated by "PD" within the text).

Lastly, I would like to thank my colleagues Roger Vogeler and Marian Anton for contributing valuable discussions that led me to revise the notes many, many times; believing in my approach; and wanting to take the journey with me.

Chapter 1

Propositional Logic

1.1 Propositions

Definition 1.1. *A proposition or statement is a declarative sentence with both a subject and predicate (verb and other modifying information) that has a unique truth value, true or false but not both, or could be assigned such a truth value.*

Example 1.2. The following are all examples of propositions:
Boston is a city in Massachusetts.
$5 \times 3 = 15$.
$5 \times 3 < 12$.

Exercise 1.3. Find the subject and verb/predicate (could be one or multiple words) of each of the propositions in example 1.2. Then determine their truth value.

Exercise 1.4. Is "5×3" a proposition? Why or why not?

We use letters, such as p and q, to denote propositions. These letters are called propositional variables.

Example 1.5. Let p be the statement "January 25 is a Tuesday" and q be "$x > 0$". Notice that in this case, we don't know if p and q are true or false without more information, but they *could* be true or false, so we still consider them propositions. It is to be noted that some contexts outside of this course would not consider these statements to be propositions until we had enough information to definitively determine their truth values.

Definition 1.6. *Let p be a proposition. The negation of p, denoted by $\neg p$, is the statement "It is not the case that p." The proposition $\neg p$ is read "not p," and the truth value of $\neg p$ is the opposite of the truth value of p.*

Exercise 1.7. In this exercise, let's focus on the fact that the truth value of the negation of *p must always have the opposite truth value* as p. For example, consider the proposition p given by "My mother was born in the U.S." Is the proposition q given by "My mother was born in Nigeria" a valid negation of p? (Could there be a situation where q and p could have the same truth value?) If q is not a valid negation of p, what could be? Try to produce a proposition that does not just insert the word "not".

Remark 1.8. In Exercise 1.7, you may wonder why we would not simply say "My mother was not born in the U.S." as our negation. While that is technically a correct negation, in general that kind of negation will not be mathematically useful to us because later, we will need to use and/or prove these negations. How do you prove that someone was not born in the U.S.? By producing a birth certificate from any other country. Similarly, how would we prove a function is say, not continuous? We can't have any idea how to do that until we have actually interpreted what it means to not be continuous.

Problem 1.9. *Negate each of the following propositions into a fluid English sentence, by restating it as what is true rather than what isn't. Your final statement should be more than just having added the word "not."*

1. *"I am under 21 years of age."*
2. *"There are fewer than 10 students in this math course who are juniors."*
3. *"At least half of the students in this math course are juniors."*
4. *"Exactly half of the students in this math course are juniors."*

Definition 1.10. *Let p and q be propositions. The* <u>*conjunction*</u> *of p and q, denoted by $p \wedge q$, is the proposition "p and q." The conjunction $p \wedge q$ is true only when p and q are BOTH true.*

Definition 1.11. *Let p and q be propositions. The* <u>*disjunction*</u> *of p and q, denoted by $p \vee q$, is the proposition "p or q." The disjunction $p \vee q$ is true when either p or q is true (false only when both p and q are false).*

Note: The disjunction "or" refers to the *inclusive or*, as opposed to the *exclusive or*. The inclusive or includes the possibility that both options may hold, as in "Cream or sugar?" The exclusive or does not include the possibility of both, as in "Fries or baked potato?"

Definition 1.12. *A* <u>*compound proposition*</u> *is one that is formed from propositional variables and logical operators (e.g. $p \wedge q, p \vee q$, etc.). Note that a compound proposition is still a proposition. Its truth value may vary depending on the truth values of the propositional variables comprising it.*

Remark 1.13. Order of Operations

So far we have learned three symbolic logical operators: $\neg, \wedge$, and $\vee$. The symbol $\neg$ takes precedence over $\wedge$ and $\vee$; for example, $\neg p \wedge q$ means $(\neg p) \wedge q$ and cannot be assumed to have the same meaning as $\neg(p \wedge q)$. [Similar to as in $-x+y$ means $(-x)+y$, not $-(x+y)$.] If we have a compound proposition that contains both $\wedge$ and $\vee$ with no parentheses, we just read left to right (as we would do with $2-1+7$ for example).

1.2 Truth Tables

We use what are called truth tables to display all the possible truth values for a given proposition or propositions. Probably the most basic is that of the negation:

p	$\neg p$
T	F
F	T

Each proposition involved in the final statement to be examined (in this case, $\neg p$) is given a column, and each possible combination of truth values of these propositions is given a row. In the cases of $p \wedge q$ and $p \vee q$, the truth table would begin as follows:

p	q	$p \wedge q$	$p \vee q$
T	T		
T	F		
F	T		
F	F		

Exercise 1.14. Complete the table started above for conjunction and disjunction based on Definitions 1.10 and 1.11.

Exercise 1.15. Use your truth tables from Exercise 1.14 to determine the truth values of the following statements:

1. Zebras are green or Washington, D.C. is the capital of the U.S.
2. Zebras are black and white or Washington, D.C. is the capital of the U.S.
3. Your math professor is 10 feet tall and $1+1=7$.
4. Zebras are black and white and Washington, D.C. is the capital of the U.S.
5. Washington, D.C. is the capital of the U.S and your math professor is 10 feet tall.

1.3 Conditional Statements

Definition 1.16. *Let p and q be propositions. The conditional statement $p \to q$ is the proposition "if p, then q". (The "then" is often left off.) p is called the hypothesis and q is called the conclusion. A conditional statement is also called an implication.*

Example 1.17. If $0 < x < 1$, then $x^2 < x$.
p: $0 < x < 1$.
q: $x^2 < x$.

Example 1.18. I'll be successful if I graduate.
p: I graduate.
q: I am successful.

Remark 1.19. What is the truth value of a conditional statement? We explore this in the following problem. Recall from Definition 1.1, a proposition cannot be both true and false. Therefore, if the statement has no contradiction to make it false, it is automatically true, even if it feels like the "truth" is an empty truth.

Problem 1.20. [1]

Your math professor Dr. Logical says that you will pass your class if you get an A on the final. In this problem, you will examine in which case(s) your professor is a liar; i.e., in which case(s) she contradicts herself. Perform the following steps to answer this:

1. *Deconstruct the conditional statement "You will pass the class if you get an A on the final" into a hypothesis p and a conclusion q.*
   p:
   q:
2. *Consider the given scenario for each line of the truth table below, and use this to fill in the third column with "T" or "F", where "False" means your professor lied (contradicted herself). Keep in mind Remark 1.19. You must justify your choices.*

p	q	$p \to q$
T	T	
T	F	
F	T	
F	F	

Remark 1.21. [2]

For those of you with a computer science/programming background, one way to think about the result of Problem 1.20 is to think about "if, then" or "for" loops. If the hypothesis of the loop isn't satisfied, does the program crash? If not, there is no "contradiction".

Problem 1.22. 1. *Use the truth table from part (2) of Problem 1.20 to determine the truth value of the following statements:*

(a) *If $f(x) = x^2$ is a continuous function, then the Nile is in Egypt.*
(b) *If $f(x) = x^2$ is a continuous function, then Homer Simpson is the President of the United States.*
(c) *If there were 365 days in the year 1776, then $3^2 = 9$.* [3]
(d) *If February has 30 days, then a liger is a real animal.*

2. *Consider the statement "If n is even, then n^2 is odd."* [4]

    (a) *Does the value $n = 5$ disprove the statement?*

    (b) *Is the statement true or false? How do you know?*

Remark 1.23. We now add the symbol $\to$ to our order-of-operations list. Our order of precedence is now:

1. $\neg$
2. $\wedge$, $\vee$
3. $\to$

Remark 1.24. There are different ways to express conditional statements. Note that what really defines a conditional statement is not simply the words "If…then…" (although that is a clear sign!), but a sense of a necessary effect, where one proposition *implies* that the other must happen or must have happened as well. This can be distinguished from a conjunction ("and"), where the two propositions are entirely independent. One is not a result of or implied by the other. So, words that could indicate a statement is a conditional are words like "necessary", "sufficient", "must", "implies", "whenever", "follows from", etc.

Example 1.25. Consider the statement "I have to buy chicken to cook dinner tomorrow." This does not have an "if…then" in it, but that does not mean it is not a conditional. Remember, a conditional statement carries a sense of dependence and/or cause-and-effect, rather than two independent actions. To test if this is a conditional statement, and if so, which part is the hypothesis and which is the conclusion, we can break it into two propositions:

r : I buy chicken.
s: I cook dinner tomorrow.

Note that we don't want to use p and q yet because we don't want to imply that one is the hypothesis and one is the conclusion: that is what we are trying to determine. So, let's try it both ways:

If I buy chicken, then I will cook dinner tomorrow.
If I cook dinner tomorrow, then I bought chicken.

(Notice we adjust the tenses accordingly to make it follow the chronological order in which the events happened.) Which one of these carries the meaning of the original? Here we must be careful: note that the original does not say that buying chicken is all I have to do to cook dinner. Maybe I have to buy carrots as well! Or, maybe I bought chicken and ended up ordering pizza because the chicken was bad. But, in the event that I did cook dinner, we know that I definitely bought the chicken because the original statement says I needed to - I can't make dinner without it. Thus, the correct formulation of this conditional statement is $s \to r$, or "If I cook dinner tomorrow, then I bought chicken."

Another way to phrase is that the hypothesis (I cook dinner tomorrow) is the sufficient condition (the one that guarantees the outcome of the other condition), and the conclusion (I bought chicken) is the necessary condition (the condition that was guaranteed by the first one).

Problem 1.26. *Rewrite the following statements in the form "If...., then....". You may change the tenses of the verbs to make your sentences make more sense. Be careful - some of them are tricky!*

1. *It's raining whenever I carry an umbrella.*
2. *It is necessary for you to get a C- in this math class to graduate.*
3. *Cheaters always lose.*
4. *We never have class on Thursday.*
5. *To get a promotion, you must wash the boss's car.*

Problem 1.27. *Translate the following sentence into symbols: "Continuity is a necessary condition for differentiability, but differentiability is a sufficient condition for continuity."*
(Note that you will have to define your own propositions here.)

1.3.1 Converse, Contrapositive, and Inverse

Definition 1.28. *Let s be the conditional statement $p \to q$. The converse of s is the statement $q \to p$, the contrapositive of s is the statement $\neg q \to \neg p$, and the inverse of s is the statement $\neg p \to \neg q$.*

Problem 1.29. *Let s be the statement "If Roger drinks coffee, Roger is alert."*

1. *Write the converse, contrapositive, and inverse of s, first in symbols and then in words. You should change the tenses where necessary to preserve the original chronological order of the events. You may also use simple negations (e.g. "not alert").*
2. *Decide from your statements in question (1) which, if any of the converse, contrapositive, and inverse follow from the original. In other words, if the original statement is true, are any of the others also necessarily true and if so, which one(s) and why? Do this without the use of a truth table, and without relying solely on truth values (meaning use the given context of Roger and his coffee.)*

3. *Confirm your answer from part (2) by creating a truth table with columns comparing the converse, contrapositive, and inverse to the original conditional statement. You must do question 2 before this one.*

Problem 1.30. [5] *Suppose you read a poem and afterward you tell your friend, "I don't usually like poetry and even I liked this poem." Your friend then responds with "That means the poem was good." Determine which of the following sentences represent(s) what your friend meant (whether or not you agree). Use the language of Definition 1.28 and the results of Problem 1.29.*

1. *If a poem is good, even people who don't like poetry will like it.*
2. *If a person who doesn't like poetry likes the poem, the poem was good.*
3. *If a person who doesn't like poetry doesn't like the poem, the poem was bad.*
4. *If a poem is bad, people who don't like poetry won't like it.*

Definition 1.31. *Two compound propositions are logically equivalent (or just equivalent) if they have all the same truth values for corresponding sets of truth values of the propositions that occur in them. The notation $p \equiv q$ denotes that p and q are logically equivalent.*

Exercise 1.32. Consider the conditional statement $p \to q$. Of the four statements: $p \to q$, its converse, its contrapositive, and its inverse, which are logically equivalent? (Hint: recall your truth table from Problem 1.29.)

Exercise 1.33. Consider the two statements from Remark 1.13, $\neg p \wedge q$ and $\neg(p \wedge q)$. Are these logically equivalent?

Exercise 1.34. This exercise is designed to help you start thinking about the next two problems in a constructive way. Consider the statement "The cat is black and fluffy."

1. Deconstruct this statement into two separate propositions separated by a logical connector.
2. What would make this statement false? (Does the cat need to be neither black nor fluffy to make this statement false?)
3. Repeat steps 1 and 2 for the statement "I'll eat hamburgers or hotdogs."

Remark 1.35. In the next problem, we are going to explore if we can "distribute" a logical negation to a compound expression with $\wedge$ and $\vee$ as in the statement $\neg(p \wedge q)$, the way we do with multiplication and addition, and if so, how? Note that we cannot *assume* we can distribute or even know how that would work, but rather we must determine that through the meaning of the expressions. This is what you will do below.

Problem 1.36. [6]

1. *Choose/define two statements p and q with parallel structure as in Exercise 1.34, and use these to write out the following compound propositions in words. Translate the statements literally as they are written (using simple negations), as opposed to interpreting them (that will be part 2).*

   $\neg(p \wedge q)$
   $\neg p \wedge \neg q$
   $\neg p \vee \neg q$
   $\neg(p \vee q)$

2. *Now use your sentences from part (1) (not a truth table) to determine which, if any, of the four given propositions are logically equivalent to each other. Make sure to justify your answer based on the context of your statements from part (1).*

Problem 1.37. [7]

Use what you have found in part (2) of Problem 1.36 to form and rephrase the negation of each statement below. (Hint: first write it symbolically as a conjunction or disjunction, using $\wedge$ or $\vee$ before negating it.)

1. *I will drink Coke or Pepsi.*
2. *x is a negative integer.*
3. *King Kong is a gorilla but Godzilla is not.*
4. *The radius or the circumference of a circle is irrational.*

Remark 1.38. The equivalences you found in Problem 1.36 are called De Morgan's Laws. Now fill them in according to what you determined in that problem to have as a reference.

$\neg(p \wedge q) \equiv$ ______________
$\neg(p \vee q) \equiv$ ______________

Problem 1.39. *Let p, q, and r be the propositions:*
p: You get an A in this math class.
q: You take every math class offered.
r: You graduate.

Write the following propositions using p, q and r and logical connectives (i.e., in symbols).

1. *You graduate, but you do not take every math class offered.*

2. *You graduate even though you don't get an A in this math class and don't take every math class offered.*
3. *Getting an A in this math class or taking every math class offered is enough to graduate.*
4. *You must get an A in this math class or take every math class offered to graduate.*

Definition 1.40. *Let p and q be propositions. The biconditional statement $p \leftrightarrow q$ is the proposition $(p \to q) \land (q \to p)$, read as "$p$ if and only if q", often abbreviated "p iff q". It can also be read as "p is necessary and sufficient for q".*

Exercise 1.41. Construct a truth table for the expression $(p \to q) \land (q \to p)$ to determine the truth values of the biconditional statement $p \leftrightarrow q$. It will be helpful to make separate columns for $p \to q$ and $q \to p$ before evaluating the final statement. Similarly, construct a truth table for $q \leftrightarrow p$. What do you notice?

Remark 1.42. Recall from Problem 1.29 that the converse of a conditional statement $p \to q$ does not necessarily hold just because the original conditional does. However, observe that with an "if and only if", if $p \leftrightarrow q$ is true, it is saying exactly what was not case before: the conditional $p \to q$ AND its converse $q \to p$ both hold.

Remark 1.43. We now add $\leftrightarrow$ to our order of precedence:

1. $\neg$
2. $\land, \lor$
3. $\to, \leftrightarrow$

1.4 Predicates and Quantifiers

Definition 1.44. *A propositional function $P(x)$ is a proposition with a variable x that can take values chosen from a set of objects, called the domain.*

Example 1.45. Let $P(x)$ be the proposition "$x < 2$". We don't know the truth value of $P(x)$ until x is assigned a value. Some possible domains might be the set of real numbers, or even just the set $\{1,2,3\}$.

Exercise 1.46. Using the propositional function $P(x)$ from Example 1.45, what are the truth values of $P(1)$ and $P(2)$?

Example 1.47. Let $P(x,y)$ be the statement "x is taller than y". Observe that P(Rihanna, Lady Gaga) has truth value "true" whereas P(Rihanna, LeBron James) has truth value "false." Observe that it is the location of the variable, not the name, that determines its role; e.g., $P(y,x)$ means "y is taller than x" since y is now in the first spot and x in the second.

Remark 1.48. A propositional function may have more than two variables, i.e. be of the form $P(x_1, x_2, \ldots, x_n)$. Some examples are, "x_1 bought lunch for x_2 and x_3", or even just "$x_1 = x_2 + x_3 + x_4$". Note that a multi-valued propositional function (e.g. $P(x,y)$) may have different domains for each variable.

Example 1.49. Let $P(x,y)$ be the statement "x has completed at least y credits at CCSU."
We choose domains for the two variables:
Domain for x: $\{$Students enrolled at CCSU$\}$
Domain for y: $\mathbb{N}$ (the natural numbers $\{0,1,2,\ldots\}$)

Exercise 1.50. Choose possible domains for the statement $P(x,y)$ given by "x loves to eat y."

We finish this section by expanding our definition of "logically equivalent" to apply to propositional functions.

Definition 1.51. *Two propositional functions $P(x)$ and $Q(x)$ are* <u>*logically equivalent*</u> *if $P(t)$ and $Q(t)$ have the same truth value for every member t of the domain, for any possible domain on which $P(x)$ and $Q(x)$ are defined.*

1.4.1 Quantifiers

Definition 1.52. *The* <u>*universal quantification*</u> *of a propositional function $P(x)$ is "$P(x)$ for all values of x in the domain." We denote it as $\forall x P(x)$, where $\forall$ is called the* <u>*universal quantifier*</u>*. We read it as "for all x $P(x)$" or "for every x $P(x)$" or "for each x $P(x)$" or "for any x $P(x)$." An element c for which $P(c)$ is false is called a* <u>*counterexample*</u> *of $\forall x P(x)$.*

Remark 1.53. Note that a proposition $P(x)$ can only consider one value of x at a time. So, you can think of $\forall x P(x)$ as a loop that runs through all values of x in the domain one at a time, testing the statement that follows on each one. If at any point $P(c)$ returns a value of "false" for any c in the domain, we have found a counterexample and we can stop, because the statement $\forall x P(x)$ is now proven false.

Exercise 1.54. Translate the sentence "Every dog loves to play" into symbols by

1. writing a statement $P(x)$ with x as the subject
2. defining an appropriate domain for x
3. using the universal quantifier $\forall$

Exercise 1.55. Let $P(x)$ be the statement $x \geq 0$. Determine if the statement $\forall x P(x)$ is true or false given the following domains (if it is false, find a *counterexample* to show that):

1. $\mathbb{N}$

2. $\mathbb{R}$

3. The solution set to the equation $x^3 + 4x^2 = -3x$.

Problem 1.56. *Let $P(x)$ be the statement "If $x > 0$ then $x^2 \geq 1$". Be sure to justify all answers to the following questions (recall that to show a $\forall$ statement is false, produce a counterexample).*

1. *What is the truth value of the statement $\forall x P(x)$, where the domain for x is $\mathbb{R}$ (the real numbers)?*

2. *What is the truth value of the statement $\forall x P(x)$, where the domain for x is $\mathbb{Z}$? ($\mathbb{Z}$ is the set of all integers, $\{\cdots -3, -2, -1, 0, 1, 2, 3 \ldots\}$)*

3. *Find another domain of x that makes the statement $\forall x P(x)$ true.*

Definition 1.57. *The existential quantification of a propositional function $P(x)$ is the proposition "There exists an element x in the domain such that $P(x)$." We denote it $\exists x P(x)$, where $\exists$ is called the existential quantifier.*

Other ways to read $\exists$:
"There is an x such that $P(x)$."
"There is at least one x such that $P(x)$."
"For some x, $P(x)$."

Remark 1.58. BE CAREFUL: $\exists x P(x)$ only guarantees the existence of ONE x such that $P(x)$. There may be more, but we are not guaranteed that.

Exercise 1.59. Let $P(x)$ be the statement "x is taller than LeBron James." What is the truth value of $\exists x P(x)$? Of $\forall x P(x)$? Note that you will have to decide which domains you are considering. Is it possible to make each one true AND false (with different domains)?

Exercise 1.60. Let $P(x)$ be the statement "$x^2 < x$". Find two different domains, one so that $\exists x P(x)$ is true and another so that $\exists x P(x)$ is false.

We extend our definition of "logically equivalent" once more.

Definition 1.61. *Two quantified statements are logically equivalent if they have the same truth values no matter what statements the propositional functions contained within represent and no matter which domains are used for each variable.*

1.4.2 Negation of Quantifiers

As with our negations in the beginning of the chapter, we will try to interpret negated quantifiers in a useful and fluid way, so that we are interpreting the statement by what *is* true rather than what *isn't*. Problem 1.64 below is designed to help you explore exactly that. However, to get warmed up, consider the following exercise, which comes from a Brazilian high school mathematics competition.

Exercise 1.62. Assume that both of the following sentences are true:

- Pinocchio always lies.
- Pinocchio says "All my hats are green."

Which of the following statements can we conclude?

1. Pinocchio has at least one hat.
2. Pinocchio has only one green hat.
3. Pinocchio has no hats.
4. Pinocchio has at least one green hat.
5. Pinocchio has no green hats.

Feel free to informally talk through this exercise. We will revisit it after Problem 1.64.

Exercise 1.63. How would you put $\neg\forall$ and $\neg\exists$ into fluid words, reading the symbols in the order they are written?

Problem 1.64. *1. Choose/define a statement $P(x)$ and write out each of the following four propositions in words, using the statement $P(x)$ you chose. Write them literally as you see them, reading each one from left to right. (Hint: keep the statement simple.)*

$\neg\exists x P(x)$
$\exists x \neg P(x)$
$\neg\forall x P(x)$
$\forall x \neg P(x)$

2. Use your sentences above to determine/illustrate which of the four propositions are logically equivalent to each other.

Problem 1.65. *Use what you have found in part (2) of Problem 1.64 to form the negation of each statement below, rephrasing to bring each negation past the quantifier (meaning the quantifier is the first thing you come to, not the negation).*

1. *All fruit are tomatoes.*
2. *Every rational number is a natural number.*
3. *There exists a discrete math student who likes purple.*
4. *There is some real number whose square is not positive.*
5. *Every person in this class has seen a movie starring William Shatner.*

Remark 1.66. The equivalences you found in Problem 1.64 are called *De Morgan's Laws for Quantifiers*. Now fill them in according to what you determined in that problem to have as a reference.

$\neg\exists x P(x) \equiv$ ______________
$\neg\forall x P(x) \equiv$ ______________

Exercise 1.67. Now go back to Exercise 1.62. You should be able to formally defend your answer now using the results from Problem 1.64.

As a reminder of why we go through the effort of rewriting/rephrasing a negation as opposed to just adding a "not" and calling it a day, let's think back to Remark 1.8 and Problem 1.9. Recall that we will ultimately be wanting to use or prove these statements. For example, how do you prove that *not all* cats are tigers? You must show there *exists* a cat that is not a tiger (a counterexample).

1.4.3 Logical Expressions Involving Quantifiers

Remark 1.68. $\forall$ and $\exists$ have precedence (in terms of order of operations) over all other logical operators except $\neg$. I.e., if we would like a quantifier to apply to two propositional functions that are joined by a logical operator, such as $P(x) \vee Q(x)$, we need parentheses: $\forall x[P(x) \vee Q(x)]$. On the other hand, the expression $\forall x \neg P(x)$ automatically means $\forall x[\neg P(x)]$ even without parentheses.

Problem 1.69. *Let $P(x)$ be the statement "x grows tomatoes" and $Q(x)$ be the statement "x grows cucumbers" where the two statements have the same domain.*

Explain the <u>difference in meaning</u> between the two sentences "Everyone grows tomatoes and cucumbers" and "Everyone who grows tomatoes grows cucumbers," and write each one symbolically using the universal quantifier $\forall$.

Problem 1.70. *Let $P(x), Q(x)$, and $R(x)$ be the following propositional functions, all with domain $\{$students in this math class$\}$.*

$P(x)$: x was born in the United States.
$Q(x)$: x speaks more than one language.
$R(x)$: x has a passport.

1. *Translate the following into fluid English sentences that demonstrate understanding:*
   (a) $\forall x[Q(x) \vee \neg R(x)]$
   (b) $\exists x[\neg P(x) \wedge (Q(x) \vee R(x))]$
   (c) $\forall x[R(x) \rightarrow (P(x) \wedge Q(x))]$. *(Hint: consider Problem 1.69)*
2. *Now translate the following statements into quantifier form:*

(a) *"There is a student in this math class who wasn't born in the U.S. and speaks more than one language, but doesn't have a passport."*

(b) *"No one in this math class was born abroad but someone in this math class has a passport."*

(c) *"Everyone in this math class who wasn't born in the U.S. speaks more than one language."*

Problem 1.71. [8]

In 2020, an infectious disease expert was heard on the local radio saying "You can't get coronavirus if someone nearby doesn't have it." Use symbolic logic and quantifiers to demonstrate how this is not what the expert really meant, by doing the following:

1. *Write in symbols what the expert said.*
2. *Describe why what they said is logically incorrect, and then correct the symbols accordingly (and translate back into words).*

1.4.4 Nested Quantifiers

Example 1.72. Let $P(x,y)$ be the statement "x has eaten y", where the domain for x is $\{$people$\}$ and the domain for y is $\{$the desserts at The Cheesecake Factory$\}$. Then the statement $\forall x \exists y P(x,y)$ is the statement "For each person x, there exists a dessert y at The Cheesecake Factory such that x has eaten y," or, more fluidly, "Everyone has eaten a dessert at The Cheesecake Factory." We know in this case that x is a person and y is a dessert by their placement in the proposition P.

Exercise 1.73. Translate the following sentence into symbols by translating it into a propositional function of two propositional variables, including domains for each variable: "Every problem in this class will be presented."

Problem 1.74. *Using $P(x,y)$ from Example 1.72, explain the difference in meaning between the following pairs of statements. (Note that this does not just mean translating the statements. You must justify why their meaning is different.) It may help to consider Remark 1.53.*

1. $\forall x \exists y P(x,y)$ *and* $\forall y \exists x P(x,y)$
2. $\forall x \exists y P(x,y)$ *and* $\exists y \forall x P(x,y)$

Problem 1.75. *Determine the truth value of the following statements, where the domain for all variables is set* $\{-3,-2,-1,0,1,2,3,4,5\}$. *Make sure to completely justify all responses.*

1. $\forall x \forall y (2x - y = 6)$.
2. $\forall x \exists y (2x - y = 6)$.
3. $\exists x \forall y (2x - y = 6)$.
4. $\exists x \exists y (2x - y = 6)$.
5. $\exists x \exists y (2x - y = 6 \wedge y = x + 1)$.

Problem 1.76. *Determine the truth value of the following statements, where the domain for all variables is* $\mathbb{R}$.

1. $\forall x \forall y (2x - y = 6)$.
2. $\forall x \exists y (2x - y = 6)$.
3. $\exists x \forall y (2x - y = 6)$.
4. $\exists x \exists y (2x - y = 6)$.
5. $\exists x \exists y (2x - y = 6 \wedge y = x + 1)$.

Remark 1.77. To negate nested quantifiers, just use DeMorgan's Laws for quantifiers to continue moving the negation inside the expression.

Exercise 1.78. Show that $p \to q \equiv \neg p \vee q$ via a truth table. (You will want this for the next problem.)

Problem 1.79. [9]

The following is a statement that comes up all the time in the subject real analysis (you may or may not remember it as the formal definition of limit from Calculus I):

We say that $\lim_{x \to a} f(x) = L$ *if*

$\forall \varepsilon \, \exists \delta \, \forall x \, (|x - a| < \delta \to |f(x) - L| < \varepsilon)$, *where the domains for* ε *and* δ *are positive real numbers and the domain for x is real numbers.*

Negate this statement, as in, $\lim_{x \to a} f(x) \neq L$ *if ..., being sure to bring the negation all the way through. (Don't worry - you are not meant to understand what it means right now. You should still be able to apply the logical rules we've learned.)*

Chapter 2

Introduction to Proofs

Note: there are several pages of reading and exercises before the first problem. Do not skip this! [10]

2.1 Evens and Odds

We start this section by considering the following statement: "The square of an odd number is odd."

Exercise 2.1. The above statement is probably a statement that seems true to you, after thinking about a few examples. Reflect for a few minutes on how you might *prove* it to be always true?

Remark 2.2. Let's note two things:

1. Even though the statement above only refers to a single number ("AN odd number"), we automatically interpret it to apply to *any* odd number. In other words, we interpret this statement as universally quantified even though it does not specifically say "for all". This will be the case throughout the chapter and the rest of the course - if the statement does not specifically imply that we are talking about a single value (such as "there exists an odd number"), we assume it to be universally quantified.

2. Hopefully as you were thinking about how to prove the statement, you realized that we really need to be able to define what "even" and "odd" actually mean.

Exercise 2.3. If you haven't already done so, take a minute to think about how you might define "even" and "odd". Then have a look at the formal definition that follows (hopefully it matches your intuition).

Definition 2.4. *The integer n is <u>even</u> if there exists an integer k such that $n = 2k$. The integer n is <u>odd</u> if there exists an integer k such that $n = 2k+1$.*

Remark 2.5. Don't get too attached to the letter "k" in Definition 2.4. It's not the name of the letter or object used that's important, but what type of object the letter represents (which is?).

Exercise 2.6. Observe that $23 = 2(11.5)$. Does this mean that 23 is even? Why or why not? Why does this not violate definition 2.4?

Exercise 2.7. Write the following integers in a form that demonstrates (via Definition 2.4) whether they are even or odd: $-7, 0, 112$.

Exercise 2.8. How could you apply Definition 2.4 to prove the statement from Exercise 2.1? You do not have to write anything official here (since you haven't learned how to yet), but brainstorm a bit.

Exercise 2.9. Use Definition 2.4 to solve the following problem algebraically: find two consecutive odd numbers whose product is 195.

Remark 2.10. What follows is some terminology we will be using throughout the rest of the course.
A *theorem* is a statement that can be demonstrated to be true (i.e. a fact or result). An *axiom* (or a *postulate*) is a statement we assume to be true. A *lemma* is a less important theorem used in the proof of another theorem. (There are others, but these are the ones we will mainly use here.)
A *proof* is a logically valid mathematical argument that establishes the truth of a statement. Note that axioms/postulates do not need to be proved (think of them as the most basic words in the dictionary that are the building blocks for other definitions).

Below are a definition and axiom upon which we will rely heavily in this chapter:

Definition 2.11. *A set of numbers is* <u>closed under an operation</u> *if performing that operation yields a result contained in the original set.*

Remark 2.12. Observe that the set of integers is closed under addition, subtraction, multiplication, meaning, the result of adding, subtracting, or multiplying integers is another integer. For example, if t is an integer, the quantity $5t^6 - 3t + 1$ is also an integer. (Note that t^6 means multiplication of six factors of t.) We will call this the *Axiom of Closure for Integers*.

Exercise 2.13.

1. Why is division not listed in the Axiom of Closure for Integers?
2. We have seen that raising an integer to a positive integer power is covered by the Axiom of Closure for Integers because it means repeated multiplication. Would the same hold for a negative integer power? (What about 0?) A fractional power?

Exercise 2.14. Under which operations are the *real numbers* (the entire number line including everything) closed? All of the ones listed in the Axiom of Closure for Integers? More?

Remark 2.15. In general, the kind of statements that we prove most often are conditional statements ($p \to q$). This means that we are trying to show that the given conditional statement can never take the truth value "False". Recall that if the hypothesis p is not satisfied, the conditional statement $p \to q$ is automatically true and there is no work to be done (the statement is vacuously true). Thus, we only really consider the case where p is satisfied, and we prove that in that case, q must also hold true. There are several main methods for accomplishing this, which we discuss in the remainder of this chapter. We start off with the most straightforward of the techniques.

2.2 Direct Proofs

1. State/assume that p is true (or simply "assume p"). p may consist of multiple propositions.
2. Make any immediate conclusions that follow from assuming p.
3. Use the conclusions from (2) to show that q is also true. (This step is usually the longest/hardest part. It can sometimes help to try to "reverse engineer", or work backward, from q.)

Remark 2.16. A few more notes: [11]

1. In <u>any</u> proof, we must always start by naming the players. I.e., if we are talking about two arbitrary real numbers r and s in the proof, we must start with the statement "Let r and s be real numbers."
2. Any time we introduce a new player, we must say what kind of object is. For example, if I say $r = 2s + t$, and I haven't already said what t is, I need to immediately do so (e.g. "for some real number t.")
3. We use "we" as opposed to "I" or "you".
4. We start each proof with <u>Proof</u>: and end each proof with either QED (quod erat demonstrandum, Latin for "which was to be demonstrated") or ■.

Exercise 2.17. Determine what is incorrect about the beginning of the proof of the following statement: n is even if $7n - 2$ is even.
<u>Proof</u>: Let n be even. Then $n = 2k$ for some integer k.......

Remark 2.18. The given definitions along with the Axiom of Closure for Integers are the only information we assume regarding even and odd numbers. Every other result about even and odd numbers we will <u>prove</u>. So, unless we have proven a given statement (or it's an axiom we have previously acknowledged), it can't be assumed. At some point, you might be wondering why we are taking the time to prove statements we already feel like we know are true. It is mainly because this is where we are learning *how* to prove, and it's much easier to learn that skill on statements we understand, and terms with which we are familiar, than on entirely

brand new concepts. Also, it reinforces that there are many things in mathematics which can't actually be taken for granted, just because they seem true. That happens all the time. Lastly, it is a fundamental mathematical principle that if we believe something to be true to the point of claiming it as fact, we should be able to prove it. It's a way of being honest with ourselves.

Exercise 2.19. Now that you have a little bit better of an idea of the structure of a proof, try Exercise 2.1 one more time on your own. Then you can turn the page to see how you did. (Hint: start by rephrasing the statement as a conditional statement to set yourself up for a direct proof.)

Example 2.20. Let's finally prove the statement from Exercise 2.1: the square of an odd number is odd.

We first rephrase the original statement as a conditional, using what we call an *arbitrary* odd number n, to "If n is an odd number, then n^2 is odd." (Note that by "arbitrary" here, we mean we pick a variable so that it could represent any number. The only additional assumption we make about n is that it is odd.)

We then move on to the steps of a direct proof.

1. Assume p and name our player(s): Let n be an odd number.
2. Draw any immediate conclusions from p: Since n is odd, $n = 2m+1$ for some integer m.
3. Show q is true too: this is the "meat" of the proof, and the part that requires the most thought. We do not have one cookie cutter method of going from p to q. However, it is always good to be constantly reminding yourself of what you **want to show** (WTS), which is in this case that n^2 is odd. We thus take a look at n^2 using the conclusion drawn in (2):

$$\begin{aligned} n^2 &= (2m+1)^2 \\ &= 4m^2+4m+1 \end{aligned}$$

Now where do we go from here? We remind ourselves what we WTS, which is that n^2 is odd, according to Definition 2.4. That means we must manipulate our expression for n^2 above so that it matches the definition of "odd", 2(integer) + 1. We note that in $4m^2+4m+1$, we already have the "$+1$", so let's separate that from the rest. That leaves us with $4m^2+4m$. Can this be written in the form of 2(integer)? Hmm.....

$$\begin{aligned} n^2 &= 4m^2+4m+1 \\ &= 2(2m^2+2m)+1 \end{aligned}$$

We're almost there! The one last thing we need to cover is to make sure that this entirely satisfies Definition 2.4, meaning that $2m^2+2m$ is not just any number but specifically an *integer* (and we know from Exercise 2.6 that this matters). So why is this the case? Thanks to the Axiom of Closure for Integers, since m itself was assumed to be an integer.

It is important to note that **this is not how we write up our final answer**; this is just the process we undergo to get there (our scrap work). We actually write our solution more efficiently, in fluid paragraph form with complete sentences as follows:

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Proof: Let n be odd. Then $n = 2m+1$ for some integer m. We have

$$\begin{aligned} n^2 &= (2m+1)^2 \\ &= 4m^2+4m+1 \\ &= 2(2m^2+2m)+1 \end{aligned}$$

Since m is an integer, so is $2m^2+2m$ by the Axiom of Closure for Integers. Thus, n^2 is odd.
■

Exercise 2.21. Find algebraic steps/manipulations that would allow you to prove the following statements, keeping both Definition 2.4 and Remark 2.5. in mind:

1. 0 is even.
2. If n is even, $n+1$ is odd.
3. If n is odd, $n+1$ is even.

Note that the three statements above combined show that every nonnegative integer must be even or odd - there can't be a nonnegative integer that is neither. (Why?)

Remark 2.22. A few final notes before you start writing your own complete proofs:

1. Sometimes, you will be asked to prove or disprove a statement. **You should always try to decide if you think it's true or false first, before you start trying to prove it.** If you believe the statement is true you must prove it. If you believe the statement is false, you should disprove it by producing a counterexample.
2. Recall that a biconditional statement $p \leftrightarrow q$ is defined as $(p \to q) \wedge (q \to p)$. I.e., two separate conditional statements must hold for the biconditional to hold. Thus, to prove a biconditional statement, we must prove both individual conditional statements separately, so it will be two proofs in one. (Question: To disprove a biconditional, do you think we need to disprove both individual conditionals?)
3. Regarding vocabulary, you can (and should) use your own words to write your proofs; they should not look like carbon copies of mine. However, there are a few words of which you must be mindful. We will call them "assumption words", "conclusion words", and "already true" or "true anyway" words.
   - Assumption words are words such as "let", "suppose", and "assume", which all imply that you are **making an assumption**, i.e. you are *choosing* to say that a statement is true.

- Conclusion words are words such as "thus", "therefore", "then", and "so", which all imply that you are **drawing a conclusion** based on previous statements.
- "Already true" words present a fact that we have either **already previously shown** in the proof, or that are **true independent of our proof**, such as "We know that $(a+b)^2 = a^2 + 2ab + b^2$", or "We have already shown that n^2 is odd."

If you misuse these words you are misstating the reasoning. For example, I may start a proof with "Let n be even", because that is an assumption I am choosing to make. However, from there, I should not say "Let $n = 2k$ for some integer k" because this *follows* from the previous statement, i.e. it is a conclusion I am drawing. I should instead say, "*Then* $n = 2k$ for some integer k," or use another conclusion word. [13]

Problem 2.23. *1. Prove or disprove: The sum of two even numbers is even.*

2. Prove or disprove: The sum of two numbers is even if and only if the two numbers are themselves even.

Problem 2.24. *1. Prove or disprove: If $2n+4$ is even then n is even.*

2. Prove or disprove: If n is even then $(n+3)^2$ is odd.

Exercise 2.25. This is a continuation of Exercise 2.21. Find algebraic steps/manipulations that would allow you to prove the following statements.

1. If n is even, $-n$ is even.
2. If n is odd, $-n$ is odd. (this one takes some creative algebra that will help you approach the next problem)

Remark 2.26. Recall that the list of results from Exercise 2.21 showed us that every nonnegative integer must be even or odd, meaning there cannot be one that's neither. Now observe that combining those results with the results of Exercise 2.25 show us that every integer (including negatives) must be even or odd, not neither. (Again, why? Also, note that the results DON'T prove that an integer can't be both even and odd simultaneously. We will prove that later, but you can assume it for now and use it freely.)

Problem 2.27. *Let n be an integer. Prove that if $n+2$ is odd, then so is n^3.*

2.3 Indirect Proofs

In general, we try the following techniques *after* attempting a direct proof. There are two types of indirect proofs, proof by contraposition and proof by contradiction. Recall from Problem 1.29 that a conditional statement and its contrapositive are equivalent. We can thus rely on that fact to prove a given conditional statement by proving its contrapositive instead.

2.3.1 Proof by Contraposition

Exercise 2.28. Let n be an integer. Try to prove the following statement using the direct proof technique that you have just learned: If n^2 is even, then n is also even. What happens?

Exercise 2.29. 1. Write the contrapositive of the statement in Example 2.28. (Note that from Exercise 2.21, Exercise 2.25, and Problem 2.41 which we will do shortly, we can conclude that "not even" means "odd" and vice versa.)

2. Prove this contrapositive statement with a direct proof.

3. Why was the additional assumption "Let n be an integer" required?

Problem 2.30. *Let n be an integer. Prove that if $n^3 - 5$ is even, then n is odd.*

Problem 2.31. *Let n be an integer. Prove that $3n+4$ is even if and only if n is even.*

Problem 2.32. *(PD) Let s,t be real numbers. Prove that if $s \neq -1$ and $t \neq -1$, then $s+t+st \neq -1$.*

Remark 2.33. You should think of a proof by contrapositive as a tool, to be used when it makes sense to use it. Generally problems don't come with a flag saying "use the contrapositive" - it is up to us to decide when it might be helpful.

2.3.2 Rational Numbers

Definition 2.34. *A real number r is rational if $r = \frac{p}{q}$ for some integers p and q with $q \neq 0$. A real number that is not rational is simply called irrational.*

Two famous irrational numbers are π and e.

Exercise 2.35. Recall that in using Definition 2.4 to show a number was odd or even, it was very important to verify that the quantity being multiplied by 2 was in fact an *integer*, not just any old number. This idea is expanded upon in the definition of rational number. What THREE things must we verify before we may conclude that Definition 2.34 is satisfied?

Problem 2.36. *1. Prove that if x is rational, then $3x$ is rational.*

2. Prove that if x and y are rational numbers, then $x+y$ is also rational.

Problem 2.37. *1. Let x be a nonzero real number. Prove that x is rational if and only if $\frac{1}{x}$ is rational.*

2. Use what you proved in part (1) to find a counterexample to the following statement: the product of two irrational numbers is irrational.

Problem 2.38. *Prove that a real number x is rational if and only if $2x+3$ is rational.*

Remark 2.39. You might be wondering why rational numbers are included specifically in the Indirect Proofs section. Note that since the definition of "irrational" is just "not rational", there is no set form of an irrational number to work with or to aim for. Thus, when you are asked to show that a number is irrational you will almost always want to use an indirect proof so that you can start off by assuming that the number is not irrational, i.e. rational.

2.3.3 Proof by Contradiction

A *proof by contradiction* is a little more interesting and freeing than a proof by contraposition. You can think of it as assuming that what we want to prove is false and showing that something "breaks" mathematically under that assumption. More formally, we can say that a contradiction is a statement that is mathematically false or impossible based on previously established mathematical definitions and knowledge. So in a proof by contradiction, we show we cannot live in a consistent mathematical world where the original given statement is false because it would lead to another mathematically false statement or contradictory information if so. We now examine the structure of a proof by contradiction.

1. Fully negate the statement you are trying to prove (so what we learned in Chapter 1 about interpreting negations will be important here). Be careful to include the "hidden" quantifier we often have in our statements.
2. Assume that the negation holds (is true).
3. Move forward from that starting assumption in direct proof fashion until you reach a contradiction of some kind, either a general mathematical contradiction such as $1 = 0$, or a contradiction of a previously-drawn conclusion.

We often indicate that we have found a contradiction by the symbol $\rightarrow\leftarrow$. We insert the symbol at the moment in the proof where we have encountered our contradiction.

Problem 2.40. [14]

1. *Write out the following statement symbolically, creating your own $P(x)$ and including an appropriate quantifier: There is no largest rational number.*
2. *Negate your statement from part (1).*
3. *Assume your negation from part (2) holds. Use this assumption to move forward into a proof by contradiction to prove the original statement.*

Problem 2.41. *Prove that an integer cannot be both even and odd.*

Problem 2.42. *Prove that the sum of a rational number and an irrational number is irrational.*

Exercise 2.43. By Problem 2.36, we know that the sum of two rational numbers is rational, and by Problem 2.42, the sum of a rational number and an irrational number is irrational. Do you think that the sum of two irrational numbers must be irrational? (Hint: "sum" also encompasses differences, by adding negative numbers.)

Problem 2.44. [15] *Prove that given any circle, its radius or its area must be irrational.*

Problem 2.45. *Prove that there is no pair of* <u>*positive*</u> *integers (x,y) that satisfy the equation $x^2 - y^2 = 1$.*

Problem 2.46. *Let $a_1, a_2, \dots, a_{10}$ be a set of nonnegative integers such that $a_1 + a_2 + \cdots + a_{10} = 40$. Prove that at least two of the numbers $a_1, a_2, \dots, a_{10}$ must be the same.*

Definition 2.47. *The rational number $\frac{m}{n}$ is* <u>*in lowest terms*</u> *if m and n have no positive common factors other than 1. (By "factor" of m, we mean an integer s such that $m = sk$ for some integer k.)*

Remark 2.48. A rational number can be specifically assumed to be in lowest terms. In a proof, we sometimes choose to do this if it will help our argument. We see this demonstrated in the following example, which is one of the most classic and well-known proofs by contradiction in mathematics.

Example 2.49. [16] Prove that $\sqrt{2}$ is irrational.
<u>Proof</u>: Suppose not, i.e. assume that $\sqrt{2}$ is rational. Then $\sqrt{2} = \frac{p}{q}$ for some integers p and q with $q \neq 0$. We may assume that the fraction $\frac{p}{q}$ is in lowest terms.

Then $2 = \frac{p^2}{q^2}$ and so $p^2 = 2q^2$.

Thus by the Axiom of Closure for Integers, p^2 is even since q^2 is an integer, because q is. By Exercise 2.29, p is even since p^2 is even. Then $p = 2s$ for some integer s, so $(2s)^2 = 2q^2$.

Then we have $4s^2 = 2q^2$ or $2s^2 = q^2$.

Thus, again by the Axiom of Closure for Integers, q^2 is even since s^2 is an integer, because s is. We see then that q is also even by Exercise 2.29, so $q = 2r$ for some integer r.

We now have that $\sqrt{2} = \frac{p}{q} = \frac{2s}{2r}$ which is not in lowest terms, which is a contradiction! $\rightarrow\leftarrow$
Thus, $\sqrt{2}$ must be irrational.
■

Problem 2.50. *Prove that $\sqrt[3]{2}$ is irrational. You may have to prove a lemma along the way. Note that this proof should be in your own words and not just copying the wording of Example 2.49.*

2.4 Concluding Thoughts

Remark 2.51. In this chapter, we have considered several types of proofs: direct proofs, proofs by contraposition, and proofs by contradiction. One more type which we will not spend too much time on here is a *proof by cases*. Note that this is not an exhaustive list of types of proofs (nor does such a list exist), but these are among the most common.

A proof by cases is a proof in which we divide the hypothesis up into two or more cases. For example, if we would like to prove something is true for all real numbers, we might have reason to consider positives, negatives, and 0 all separately. This is just an example of a set of possible cases; what the cases are and how many there are of them depends on the statement to be proved.

Problem 2.52. [17] *Prove that if n is an integer, then n^2+5n+7 is odd.*

Lastly, this next problem is in here just for fun.

Problem 2.53. [18] *Find the mistakes in the following "proofs" (the lines are numbered just to make it easier for you to refer to them):*

Claim: $2=1$:
Proof: Suppose that r and s are two real numbers, with $r=s$. Then we have the following:

1. $r=s$
2. $r^2=rs$ *multiplied both sides by r*
3. $r^2-s^2=rs-s^2$ *subtracted s^2 from both sides*
4. $(r-s)(r+s)=s(r-s)$ *factored*
5. $r+s=s$ *canceled $(r-s)$*
6. $2s=s$ *substituted r in for s since they are equal by assumption*
7. $2=1$ *canceled s*

■

Claim: $1=0$:
Proof:

1. *Observe that $1=x\left(\frac{1}{x}\right)$ for $x\neq 0$*

2. *Take the derivative of the right side using the product rule and simplify:*
   $1\left(\frac{1}{x}\right) + x\left(-\frac{1}{x^2}\right) = \frac{1}{x} - \frac{1}{x}$

3. *Integrate what we just found as the derivative of the right side, which brings us back to our original equation:*
   $1 = \int [\frac{1}{x} - \frac{1}{x}]\, dx = \int \frac{1}{x}\, dx - \int \frac{1}{x}\, dx$

4. *Bring the* $-\int \frac{1}{x}\, dx$ *to the left side:* $1 + \int \frac{1}{x}\, dx = \int \frac{1}{x}\, dx$

5. *Cancel the* $\int \frac{1}{x}\, dx$*:* $1 = 0$

Chapter 3

Set Theory

3.1 Introduction to Sets

Definition 3.1. *A set is an unordered collection of "objects". The objects in a set are called the elements or members of the set. A set contains its elements. A set is usually denoted by a capital letter, say A, and an element is usually denoted by a lowercase letter, say a or b. We write $a \in A$ to mean "a is an element of A". $a \notin A$ means the statement $\neg(a \in A)$, or "a is not an element of A".*

Definition 3.2. *We can write sets in roster notation, which is simply a list of the elements, or in set-builder notation, which describes a defining characteristic common to all the elements of the set. In either case, we encase the list or description in curly brackets $\{\}$. In roster notation, if the set is infinite or if there is a long list where the pattern is clear, we can indicate this with "…". Set-builder notation is set up as follows:*
{kind or name of object : specific description} or
{kind or name of object | specific description}.
We read the : or | as "such that".

Example 3.3. $A = \{\text{red, yellow, blue}\}$ is a set in roster notation, and an equivalent description in set-builder notation is $A = \{x \text{ is a color} : x \text{ is primary}\}$ or even $A = \{x : x \text{ is a primary color}\}$. We read the second one as, "x is a color such that x is primary" and the third as "x such that x is a primary color."

Example 3.4. $\{a, \text{dog}, \text{math}, -2\}$ is a set which is equal to $\{-\frac{4}{2}, \text{dog}, a, \text{math}\}$ and $\{a, \text{dog}, a, -\frac{6}{3}, \text{math}, -2\}$ but not $\{a, \text{dog}, \text{math}, -1\}$. In other words, order does not matter nor does number of appearances of an element, nor the manner in which it is expressed (as long as it is indeed the same element). Note that in this case, we would probably not attempt to use set-builder notation since it would be difficult to find a defining property common to all of these elements.

Below are some commonly-encountered sets (note no curly brackets when we use the special symbols):

$\mathbb{R}$: the real numbers
$\mathbb{Z}$: the integers
$\mathbb{N}$: the natural numbers $\{0,1,2,\dots\}$ (Note: in some settings $\mathbb{N}$ does not include 0)
$\mathbb{Z}^+$: the positive integers $\{1,2,3,\dots\}$
$\mathbb{Q}$: the rational numbers
$\mathbb{C}$: the complex numbers

Using this notation, we can now say "x is a real number", for example, by simply writing $x \in \mathbb{R}$. With this in mind, note that the set $\mathbb{C}$ can be described in set-builder notation as $\{a+bi : a,b \in \mathbb{R}, i = \sqrt{-1}\}$.

Exercise 3.5. For each of the following sets, write out at least five elements. Then determine which ones are the same.

1. $\{4m : m \in \mathbb{Z}\}$
2. $\{4t : t \in \mathbb{Z}\}$
3. $\{n+4 : n \in \mathbb{Z}\}$
4. $\{r : r = 4k \text{ for some } k \in \mathbb{Z}\}$
5. $\{4x+7 : x \in \mathbb{Z}\}$
6. $\{4s+8 : s \in \mathbb{Z}\}$

Exercise 3.6. Describe the set of even numbers, the set of odd numbers, and the set $\mathbb{Q}$ in set-builder notation.

Problem 3.7. [19]

Write the following sets in roster notation.

1. $\{x \in \mathbb{N} : x = 2s \textit{ for some } s \in \mathbb{Z}, 3 \leq x < 10\}$
2. $\{2s-1 : s \in \mathbb{Z}, 3 \leq s < 10\}$
3. $\{x \in \mathbb{Z}^+ : x \textit{ is prime and } x \leq 30\}$
4. $\{x \in \mathbb{R} \mid x^3+4x = 0\}$
5. $\{z \in \mathbb{C} \mid z^3+4z = 0\}$

Problem 3.8. *Write the following sets in set-builder notation.*

1. $\{2,4,6\}$
2. $\{0,3,6,9\}$
3. $\{0,1,16,81,256,625,1296,\dots\}$
4. $\{\dots,100000,10000,1000,100,10,1,.1,.01,.001,.0001,\dots\}$
5. $\{\dots,-8,-2,4,10,16,\dots\}$

Definition 3.9. *A set containing exactly one element is called a singleton set.*

Definition 3.10. *The set with no elements is called the empty set, denoted $\emptyset$ or $\{\ \}$ (but not $\{\emptyset\}$).*

Definition 3.11. [20] *The set A is a subset of the set B if $\forall x(x \in A \to x \in B)$; we denote this as $A \subseteq B$.*

Exercise 3.12. Write Definition 3.11 in words.

Problem 3.13. *Show that for every set S the following are true, by justifying (using the formal logic that we learned in Chapter 1) why the defining statement in Definition 3.11 is always true for each one.*

1. $\emptyset \subseteq S$
2. $S \subseteq S$

Definition 3.14. *Given a set S, the power set of S is the set of all subsets of the set S. We denote this $P(S)$.*

Exercise 3.15.

1. List all the subsets of the set $S = \{\pi, 3, 0\}$.
2. Now list all your sets from part (1) within another set. This is the power set $P(S)$.

While it may seem confusing to see a set within a set, you might think of each set as a bag. A former student once described it as a bag of Halloween candy: the big bag might contain a bag of Skittles, but not the individual Skittles themselves. So when you reach in the bag of candy, you can't access an individual Skittle (or mix it with an M&M from another bag).

Exercise 3.16. What is the difference between $\emptyset$ and $\{\emptyset\}$? You may use the bag analogy.

Question 3.17. Why are $\emptyset$ and S always elements of $P(S)$ for any set S? Is $\{\emptyset\}$ always an element in $P(S)$?

Exercise 3.18. Is $b \in \{a, \{b, c\}\}$? Is $\{b\} \subseteq \{a, \{b, c\}\}$? (Go back to the definition of subset if you are not sure.)

Problem 3.19. *Find the power sets of the following sets.*

1. $S = \{a, bagel, \$\}$
2. $T = \{a, \emptyset\}$
3. $U = \{a, \{a\}\}$
4. $V = \{a, \{a, \emptyset\}\}$

Problem 3.20. *(PD) Complete each statement, if possible, with $\in$ or $\subseteq$. (Recall that $[0,1]$ is the closed interval in $\mathbb{R}$ from 0 to 1, $\{x \in \mathbb{R} : 0 \leq x \leq 1\}$.) Be sure to justify each choice.*

1. 0.5 ___ $[0,1]$
2. $\{0.5\}$ ___ $[0,1]$
3. $\{0.5\}$ ___ $P([0,1])$
4. 0.5 ___ $P([0,1])$
5. 0.5 ___ $\{0,1\}$
6. $\emptyset$ ___ $\{0,1\}$
7. $\{\emptyset\}$ ___ $\{0,1\}$

Problem 3.21. [21] *Find sets A and B such that $A \subseteq B$ and $A \in B$ both hold.*

3.1.1 Set Operations

At this point you probably have an informal idea what it means to say that two sets are the same set; i.e., that they are equal. The first definition in this section formalizes this.

Definition 3.22. [22] *Sets A and B are equal if $\forall x(x \in A \leftrightarrow x \in B)$. Note that this is equivalent to $\forall x[(x \in A \rightarrow x \in B)$ and $(x \in B \rightarrow x \in A)]$*

Remark 3.23. Note that Definition 3.22 is equivalent to: $A = B$ if and only if $A \subseteq B$ and $B \subseteq A$. We often use this interpretation to prove that sets are equal.

Definition 3.24. *Let A and B be sets. The union of the sets A and B, denoted by $A \cup B$, is the set that contains those elements that are either in A or B or both. That is,*

$$A \cup B = \{x : x \in A \text{ or } x \in B\}$$

Definition 3.25. *Let A and B be sets. The intersection of the sets A and B, denoted by $A \cap B$, is the set containing those elements in both A and B. That is,*

$$A \cap B = \{x : x \in A \text{ and } x \in B\}$$

Definition 3.26. *Let A and B be sets. The difference of A and B, denoted by $A - B$, is the set containing those elements that are in A but not in B. That is,*

$$A - B = \{x : x \in A \text{ and } x \notin B\}$$

$A - B$ is also called the complement of B in A or the complement of B with respect to A. (Note: $A - B$ is denoted $A \setminus B$ in some settings.)

Exercise 3.27. Suppose that for sets A and B, $A \cup B = \{0,1,2,3,4,5,8,9\}$, $A - B = \{1,4,8\}$, and $B - A = \{0,2,5\}$. Find $A \cap B$.

Exercise 3.28. When is it true (if ever) that $A \cup B = A \cap B$?

Definition 3.29. *Let A and B be sets. The Cartesian product of A and B, denoted by $A \times B$, is the set of all ordered pairs (a,b) where $a \in A$ and $b \in B$. That is,*

$$A \times B = \{(a,b) : a \in A \text{ and } b \in B\}$$

Note that by "ordered pair", we mean that order matters. I.e., $(a,b) \neq (b,a)$ unless $a = b$.

Exercise 3.30. Let $A = \{0,1\}$ and $B = \{\text{dog, cat, mouse}\}$. Compute the sets $A \times B$, $B \times A$, $A \times A$, and $B \times B$ (in roster notation).

Exercise 3.31. Let $A = \{0,1\}$ and $B = \emptyset$. Compute the set $A \times B$. Do you think this is the same as $B \times A$?

Problem 3.32. *(PD)* [23]

1. *Find nonempty sets A,B,C,D such that $(A-B) \times (C-D) = (A \times C) - (B \times D)$.*
2. *Is it always true that $(A-B) \times (C-D) = (A \times C) - (B \times D)$, for all nonempty sets A,B,C,D contained within a universe U?*

Definition 3.33. *Let S be a finite set. The cardinality of S, denoted $|S|$, is the number of distinct elements in S.*

Remark 3.34. Note that cardinality can still be defined for infinite sets but it is trickier because there are different-sized infinities. (Yup! E.g. there are "more" real numbers than integers.)

Exercise 3.35. Are the following statements always true? If not, under what circumstances are they true (if ever)?

1. $|A-B| = |A| - |B|$
2. $|A-B| = |B-A|$

Exercise 3.36. If A and B are two finite sets, find a formula for $|A \cup B|$ in terms of A and B. (Careful: remember that A and B may have some elements in common.)

Definition 3.37. *Given a collection of elements and sets, the universal set (or universe) is the larger set containing these elements and sets.*

Definition 3.38. *Let U be the universal set. Then the complement of the set A, denoted $\bar{A}$, is the set $U - A$. Alternatively, $\bar{A} = \{x : x \notin A\}$. We can observe that $\forall x (x \in \bar{A} \leftrightarrow x \notin A)$.*
Note that you must know what the universe U is for $\bar{A}$ to be defined.

Problem 3.39. *Let $U = \{0,1,2,\ldots,10,11\}$ (universe), $A = \{0,3,7\}$ and $B = \{x \in U : x = 5k$ for some $k \in \mathbb{Z}\}$.*

1. *Find $|A \cup B|, |A \cap B|, |A - B|, |B - A|$, and $|A \times B|$.*
2. *Find $|\overline{A \cup B}|, |\overline{A \cap B}|, |\overline{A - B}|, |\overline{B - A}|$, and $|\overline{A \times B}|$. (Note: the universe for $A \times B$ is $U \times U$.)*

Definition 3.40. *A set A is a* <u>*proper subset*</u> *of a set B, denoted $A \subset B$, if*

$$\forall x(x \in A \to x \in B) \text{ and } \exists x(x \in B \text{ and } x \notin A).$$

Exercise 3.41.
1. The first part of Definition 3.40 should be familiar to you: what is that the definition of?
2. Interpret the second part of Definition 3.40.

Remark 3.42. The proper subset notation should never be used unless you are sure that equality is not possible (as with " $<$ " - one should never write $x < y$ unless it is certain that $x \neq y$).

Exercise 3.43. Write out in (fluid) words, the logical expression $\forall x(x \in A \to x \in B)$ and $\exists x(x \in B$ and $x \notin A)$ from Definition 3.40.

Problem 3.44. *Consider the following sets, assuming that the universe of discourse $U = \{$U.S. residents who are capable of speaking a language$\}$:*

$A = \{x \in U : x$ *speaks English*$\}$
$B = \{x \in U : x$ *speaks both English and Spanish*$\}$
$C = \{x \in U : x$ *speaks Spanish*$\}$
$D = \{x \in U : x$ *speaks only Spanish*$\}$

*Determine if the statements below are true or false (and explain). You may use your knowledge of the real world. (***Hint: keep in mind the** <u>**definition**</u> **of** $A \subseteq B$.*)*[24]

1. $A \subseteq B$
2. $B \subseteq A$
3. $B \subset A$
4. $A \cup C = B$
5. $A \cap D = \emptyset$
6. $A \subseteq \overline{D}$
7. $B - A = C$

Exercise 3.45. [25] Let $E = \{2k : k \in \mathbb{Z}\}$ and let $O = \{2l + 1 : l \in \mathbb{Z}\}$. Restate the following using notation that uses E and O: if n is even then $n^2 + 1$ is odd.

Exercise 3.46. [26]

1. Write out a few elements of each set, and determine all subset relationships among the sets (including if the containment is proper in particular).

$$A = \{4t - 8 : t \in \mathbb{Z}\} \qquad C = \{(2l)^2 : l \in \mathbb{Z}\}$$
$$B = \{16p : p \in \mathbb{Z}\} \qquad D = \{2q + 46 : q \in \mathbb{Z}\}$$

2. In each subset relationship you found, write out the corresponding conditional statement. If the containment is proper, write out the corresponding additional statement.

We are now about to move on to *proofs* of statements like your conclusions above. The next problem is meant to help you structure such a proof.

Problem 3.47. *Let* $A = \{9k^4 : k \in \mathbb{Z}\}$ *and let* $B = \{m^2 : m \in \mathbb{Z}\}$.

1. *Write out a few elements of sets A and B. (This is not part of the proof, but just to make sure you understand what type of elements each set contains.)*
2. *Consider the statement* $A \subseteq B$. *Write out the corresponding conditional statement.*
3. *Determine the hypothesis and conclusion of your conditional statement.*
4. *Assume the hypothesis (as we do in all direct proofs). What is an immediate conclusion you can make based on this assumption?*
5. *Finish the proof of the statement by moving forward from where you left off in (4).*

Problem 3.48. *Prove or disprove the following statements (it may help to give names to each set in question):*

1. $\{2s : s \in \mathbb{Z}\} \subseteq \{t^2 : t \in \mathbb{Z}\}$
2. $\{4k : k \in \mathbb{Z}\} \subset \{2l : l \in \mathbb{Z}\}$ *(note:* $\subset$*, not just* $\subseteq$*)*

Problem 3.49. *Prove that* $\{3r + 1 : r \in \mathbb{Z}\} = \{3k - 2 : k \in \mathbb{Z}\}$ *(keep in mind Definition 3.22 and Remark 3.23).*

Exercise 3.50. Are the sets $\{2m : m \in \mathbb{Z}\}$ and $\{2m + 2 : m \in \mathbb{Z}\}$ the same set? In other words, in Problems 3.47, 3.48, and 3.49, did the letters all have to be different in each problem, in defining the sets?

Problem 3.51. *Prove or disprove the following statements:*

1. $\{6t : t \in \mathbb{Z}\} \subseteq \{2t : t \in \mathbb{Z}\} \cup \{3t : t \in \mathbb{Z}\}$

2. $\{2n : n \in \mathbb{Z}\} \cup \{3n : n \in \mathbb{Z}\} = \{6n : n \in \mathbb{Z}\}$

3. $\{2r : r \in \mathbb{Z}\} \cap \{3r : r \in \mathbb{Z}\} = \emptyset$

Example 3.52. This is actually an exercise and an example in one. We are now moving on to proving more general statements. For example, consider the statement $\overline{A \cap B} = \bar{A} \cup \bar{B}$. Recall that to prove this, we should show that

$$\overline{A \cap B} \subseteq \bar{A} \cup \bar{B} \qquad \text{and} \qquad \bar{A} \cup \bar{B} \subseteq \overline{A \cap B}.$$

Start with the first one: $\overline{A \cap B} \subseteq \bar{A} \cup \bar{B}$

1. Write the statement $\overline{A \cap B} \subseteq \bar{A} \cup \bar{B}$ as a conditional statement.
2. Assume the hypothesis of that conditional statement.
3. Interpret the definition of set complement to make a conclusion.
4. Interpret the definition of intersection to make a further conclusion.
5. At this point, we can say that we have "unpacked" the hypothesis as much as we can, so now we have to figure out how to "repack" the information in such a way as to allow us to reach the conclusion. Try to complete the proof from here before turning the page.

Hopefully you came up with something like this (not necessarily in these exact words):

Proof: Let $x \in \overline{A \cap B}$. Then $x \notin A \cap B$ by definition of complement. This means that, by definition of intersection, x is not in both sets A and B simultaneously. Therefore $x \notin A$ or $x \notin B$. (Note that we have actually used DeMorgan's law here.)

Thus, by definition of complement, since $x \notin A$ or $x \notin B$, we have that $x \in \bar{A}$ or $x \in \bar{B}$.

Finally by definition of union, this yields that $x \in \bar{A} \cup \bar{B}$. Thus, $\overline{A \cap B} \subseteq \bar{A} \cup \bar{B}$.

Showing the set inclusion the other direction will take us in a slightly different direction, as we will see below. This is because we start off by assuming $x \in \bar{A} \cup \bar{B}$, which means $x \in \bar{A}$ *or* $x \in \bar{B}$. Having an "or" means we cannot definitively conclude that x is an element of either or both sets, but we can consider the different possibilities and see where each one leads us. This suggests a proof by cases, as you might remember from Chapter 2. The two cases we will consider are $x \in \bar{A}$ or $x \notin \bar{A}$. You might wonder why we wouldn't consider $x \in \bar{A}$ or $x \in \bar{B}$ as our cases. That wouldn't technically be wrong, but those cases could technically be overlapping since x could be an element of both sets. It's thus a bit "cleaner" to consider two clearly non-overlapping cases: x is either an element of $\bar{A}$ or it isn't.

Let $x \in \bar{A} \cup \bar{B}$. Thus, $x \in \bar{A}$ or $x \in \bar{B}$ by definition of union. We consider two possible cases: $x \in \bar{A}$ or $x \notin \bar{A}$.

Suppose first that $x \in \bar{A}$. Then $x \notin A$ by definition of complement, so $x \notin A \cap B$ by definition of intersection. This yields that $x \in \overline{A \cap B}$.

Now suppose that $x \notin \bar{A}$, which means that $x \in \bar{B}$ since we previously concluded that $x \in \bar{A}$ or $x \in \bar{B}$. Similarly, we then have that $x \notin B$ by definition of complement, so $x \notin A \cap B$ again by definition of intersection. This also yields that $x \in \overline{A \cap B}$.

Thus, in either case, $x \in \overline{A \cap B}$. We have thus shown that $\bar{A} \cup \bar{B} \subseteq \overline{A \cap B}$. Putting this together with $\overline{A \cap B} \subseteq \bar{A} \cup \bar{B}$, which we proved above, we have that $\overline{A \cap B} = \bar{A} \cup \bar{B}$.
■

Remark 3.53. The type of proof in the previous example is sometimes called "element-chasing".

Example 3.54. Prove the "distributive law", $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.

We once again show this by showing that

$$A \cup (B \cap C) \subseteq (A \cup B) \cap (A \cup C) \quad \text{and} \quad (A \cup B) \cap (A \cup C) \subseteq A \cup (B \cap C).$$

Try to prove the first direction on your own before turning the page. Note that your starting hypothesis will be $x \in A \cup (B \cap C)$, which will split into $x \in A$ or $x \in B \cap C$, which again leads to two cases as in the previous example.

Proof:
$A \cup (B \cap C) \subseteq (A \cup B) \cap (A \cup C)$:

Let $x \in A \cup (B \cap C)$. Then $x \in A$ or $x \in B \cap C$ by definition of union. We will consider two cases: $x \in A$ and $x \notin A$.

First suppose that $x \in A$. Then $x \in A \cup B$ and $x \in A \cup C$ by definition of union. Thus, $x \in (A \cup B) \cap (A \cup C)$ by definition of intersection.

Now suppose that $x \notin A$. Then $x \in B \cap C$ since it was already determined that $x \in A$ or $x \in B \cap C$. Thus $x \in B$ and $x \in C$ by definition of intersection. Then, by definition of union, $x \in A \cup B$ since $x \in B$, and $x \in A \cup C$ since $x \in C$. Thus, $x \in (A \cup B) \cap (A \cup C)$ by definition of $\cap$.

This shows that $A \cup (B \cap C) \subseteq (A \cup B) \cap (A \cup C)$.

The other direction of the set containment is left for the following problem.

Problem 3.55. *Show the other set containment from Example 3.54 (which will prove the set equality $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$).*

Problem 3.56. *Prove that $(B - A) \cup (C - A) = (B \cup C) - A$.*

Problem 3.57. *Prove that $(A - B) \times (C - D) \subseteq (A \times C) - (B \times D)$. (Hint: recall that an arbitrary element in $(A - B) \times (C - D)$ is an ordered pair.)*

Problem 3.58. *(PD) Prove that for sets A and B, if $A \cup B = A \cap B$ then $A = B$.*

Problem 3.59. *Prove that $P(A) \subseteq P(B)$ if and only if $A \subseteq B$.*

Definition 3.60. *Two sets are called disjoint if their intersection is the empty set. That is, A and B are disjoint if $A \cap B = \emptyset$.*

Remark 3.61. Note that to show two sets are disjoint, we must show their intersection contains no elements; i.e., there does not exist an element in their intersection. Since it can be hard to directly show that an object does *not* have a certain property or does *not* exist, *an indirect proof can often be easier*.

Exercise 3.62. Let $A = \{2k : k \in \mathbb{Z}, 0 \leq k < 10\}$ and $B = \{x \in \mathbb{Z}^+ : x \leq 7\}$. Are A and B disjoint? Are $A - B$ and $B - A$ disjoint?

Problem 3.63. [27] *Prove or disprove: For any two sets A and B, $A - B$ and $B - A$ are disjoint.*

Problem 3.64. *Prove that if $A \subseteq B \cap C$, then A and $\overline{B \cup C}$ are disjoint.*

Exercise 3.65. If the sets A and B are disjoint, what can we say about the sets $A - B$ and $B - A$?

Problem 3.66. *Prove that* $(A-B)\cup(B-A)=A\cup B$ *if and only if* A *and* B *are disjoint.*

Problem 3.67. [28] *Find a sufficient condition for* $(A-B)\times(C-D)=(A\times C)-(B\times D)$ *to hold, and prove it.*

Problem 3.68. [29] *Find a necessary condition for* $(A-B)\times(C-D)=(A\times C)-(B\times D)$ *to hold, and prove it.*

Chapter 4

Functions

4.1 Existence and Uniqueness Proofs

Definition 4.1. *An existence proof is a proof of a proposition of the form* $\exists x P(x)$.

Remark 4.2. An existence proof is very different from what we have been doing up until now. So far, we have been proving universally quantified statements, i.e., a statement that holds *for all* members x of a certain set (e.g. every even number, or every rational number). Existence is much less stringent, because we only need one. So, even though all along we have been saying that we cannot prove a statement with one example, finally here is a kind of statement where we can do exactly that!

Example 4.3. Consider the statement "There is a student at this school who passed Calculus II." To prove this statement, you would first produce the student, and then offer proof that the student has indeed passed Calculus II (transcripts, etc.) Note that just producing the student and offering no further evidence is not enough, because you would be asking everyone to take your word for it that the student indeed passed the course. So, transcripts or some other evidence of passing would also be required. Another important thing to note: you do not need to (and therefore should not) explain *how* you found that student to prove that it's true. In other words, explaining that you asked all your friends and classmates until you found someone who took Calculus II is superfluous and thus does not belong in the proof. All you do is produce the student along with proof that they passed Calculus II.

Example 4.4. Show that there exist real numbers x and y of opposite sign such that $x^2 - y^2 = 1$.
Proof: Let $x = 3$ and $y = -\sqrt{8}$.
We have that $3^2 - (-\sqrt{8})^2 = 9 - 8 = 1$. Also, observe that 3 is positive and $-\sqrt{8}$ is negative. Thus, there exist $x, y \in \mathbb{R}$ of opposite sign such that $x^2 - y^2 = 1$. ■

Note that we simply produced the objects and then showed that they had the desired properties; we did not show *how* we found those values of x and y. Though this seems strange, to not "show our work", we don't do it because *logically*, knowing how we got the values doesn't add anything more.

Problem 4.5. *Prove that there exists a function f that is continuous on $\mathbb{R}$, such that $f'(x) = 3x^4 - 10x^2 + 2x + 1$ and $f(-1) = 3$.*

Problem 4.6. [30]

1. *Let $2\mathbb{Z}$ denote the set of even integers. Prove that there exists an $x \in 2\mathbb{Z}$ such that $\frac{x-10}{2} = 33$.*
2. *Prove that for every $y \in \mathbb{Z}$, there exists an $x \in 2\mathbb{Z}$ such that $\frac{x-10}{2} = y$.*

Remark 4.7. Note that in existence proofs, we are showing that *at least* one object exists that satisfies the requirements, but we are making no claim about how many other objects might do the same. For example, in Example 4.4, you can probably find another pair x and y that would satisfy the requirements. Now, we will consider the situation where want to show that *at most* one object can satisfy the requirements. These are called "uniqueness proofs". Note that the mathematical sense of the word "unique" is "only one", as opposed to our common use of it to mean "unlike the others".

Definition 4.8. *A uniqueness proof is a proof of a statement of the form "If there exists an x such that $P(x)$, then x is unique".*

Remark 4.9. We have a standard way to prove uniqueness. In general, to prove that at most one x can satisfy the statement $P(x)$, we use the following basic procedure:

1. Suppose that there exist, say x_1 and x_2 such that $P(x_1)$ and $P(x_2)$ hold.
2. Show that $x_1 = x_2$.

Or, if we have already found one x_0 such that $P(x_0)$ holds, we can also

1. Suppose we have x_1 such that $P(x_1)$ holds.
2. Show that x_1 must be equal to the original x_0 we already found.

Note that either way, we are showing that at most one value will do the job. (Note that the first approach does not show that at least one such value exists. It just shows that *if* such a value does exist, there cannot be another one.)

Example 4.10. Show that there exists a unique x such that $3x^3 - 2 = 5$.

(First, note that that this is an existence AND uniqueness proof in one, so we must prove both.) Proof:
Existence
Observe that $x = \sqrt[3]{\frac{7}{3}}$ yields $3\left(\sqrt[3]{\frac{7}{3}}\right)^3 - 2 = 3(\frac{7}{3}) - 2 = 7 - 2 = 5$.

Uniqueness
Suppose that there exist x_1 and x_2 such that $3x_1^3 - 2 = 5$ and $3x_2^3 - 2 = 5$. Then
$3x_1^3 - 2 = 3x_2^3 - 2$
$3x_1^3 = 3x_2^3$
$x_1^3 = x_2^3$
$x_1 = x_2$.
Thus, there exists a unique x such that $3x^3 - 2 = 5$. ■

Exercise 4.11. Where does the above proof of uniqueness fall apart if instead we had the equation $3x^2 - 2 = 5$? Does it make sense that it does?

Problem 4.12. [31] *Prove that there exists exactly one additive identity for the real numbers. (Recall that an additive identity is a number h such that $h + a = a + h = a$ for every real number a.)*

Exercise 4.13. Is it necessarily true that for sets A, B, C, if $A \cup B = A \cup C$, then $B = C$?

Problem 4.14. [32] *Prove that there exists a unique set A such that $A \cup B = B$ for every set B. (Hint: note that B can be any set here.)*

4.2 Functions - Basic Definitions

In this section, we will define the concept of function more formally than how you have probably thought of it in the past. (We aren't giving it a *different* meaning, but rather a more generalized form that applies to all sorts of functions, including those you know from precalculus and calculus.)

Definition 4.15. *Let A and B be nonempty sets. A function f from A to B is a subset of the Cartesian product $A \times B$ such that there is a unique ordered pair corresponding to each $a \in A$. We write $f : A \to B$ to indicate that f is from A to B, and $f(a) = b$ to indicate that the ordered pair (a, b) is an element of f. Functions are also often called mappings or transformations depending on the context.*

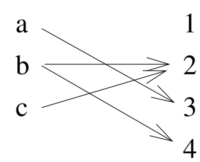
The definition above does correspond to our more familiar definition of "function": a way of assigning to each "input" element of A exactly one "output" element of B. In the Cartesian product representation of a function, the first element of the ordered pair can be thought of as the "input" element and the second element of the ordered pair can be thought of as the "output" element, just as you are used to from algebra and precalculus when graphing functions in the Cartesian plane. See the following example.

Example 4.16. Let $A = \{a, b, c\}$ and $B = \{1, 2, 3, 4\}$. Then
$A \times B = \{(a,1),(a,2),(a,3),(a,4),(b,1),(b,2),(b,3),(b,4),(c,1),(c,2),(c,3),(c,4)\}$.
We consider f as a subset of $A \times B$, say, $f = \{(a,3),(b,2),(b,4),(c,2)\}$. But, this does not represent a function because $f(b) = 2$ and $f(b) = 4$ and according to the

definition of function, there must be *exactly one* pair for each element of A. Also, the subset of $A \times B$ given by $g = \{(a,2),(c,3)\}$ is not a function from A to B because *each* element of A must have a corresponding ordered pair. However, the subset $h = \{(a,3),(b,2),(c,2)\}$ IS a function from A to B because each element of A has exactly one pair.

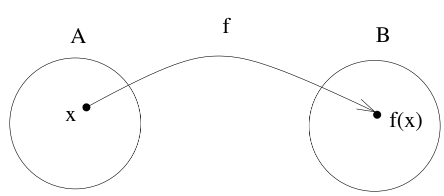
Remark 4.17. We can still use this definition/notation of function on the functions we know and remember from precalculus, such as $f(x) = x^2$. In that case we say $f : \mathbb{R} \to \mathbb{R}$ is given by $f = \{(x,x^2) : x \in \mathbb{R}\}$. We see that $A \times B$ is simply $\mathbb{R} \times \mathbb{R}$ here, otherwise known as $\mathbb{R}^2$, or the Cartesian (xy) plane.

Remark 4.18. It is often helpful to view A and B as two columns, if they do not have too many elements. We can represent f from Example 4.16 as follows:



In this way, we can easily see that f is not a function because b is assigned more than one element from B.

If A and B have many elements (or are not explicitly defined), we use a more general type of diagram:



Definition 4.19. *Let A and B be sets. If f is a function from A to B, we say that A is the domain of f and B is the codomain of f. If $f(a) = b$ we say that b is the image of a and a is a pre-image of b. The range of f is the set of all images of elements of A, i.e. the set $\{b \in B : b = f(a)$ for some $a \in A\}$. Lastly, we say that f maps A to B.*

Remark 4.20. Given the terms in Definition 4.19, from here on out we should avoid using the words "input" and "output" but rather instead use "pre-image" and "image".

Exercise 4.21. Look back at the function h defined in Example 4.16. Identify the domain, codomain, range. How many images does each element of the domain have? How many pre-images does each element of the codomain have?

Question 4.22. Why, in Definition 4.19, do we say "b is **the** image of a and a is **a** pre-image of b"?

Question 4.23. What is the difference between the codomain and the range?

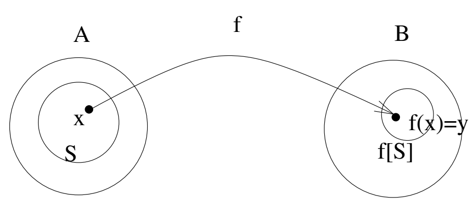
Exercise 4.24. Let f be the function that maps a student in this class to his/her age in years. (Why is this a function?) Determine the domain of this function, and a possible codomain and range. Then find the images of your group members. Remember to use correct terminology.

Remark 4.25. Note that in Exercise 4.24, you were asked to find the set of images of just your group members, even though the domain was the entire class. In other words, a specific subset of the domain was chosen, and just the images of that subset were considered. We formalize this in the following definition.

Definition 4.26. *Let A and B be sets. Let $f : A \to B$ be a function and let $S \subseteq A$. The image of S under the function f is the subset of B that consists of the images of the elements of S. We denote the image of the subset S by $f[S]$. We may also write the definition symbolically:*

$$f[S] = \{t \in B : \textit{there exists } s \in S \textit{ such that } t = f(s)\}$$

Revisiting our diagram, the image of the set S can be interpreted as below, where $x \in S$, so $f(x) = y \in f[S]$. Note that in this diagram we are assuming S is a *proper* subset of A, though it does not have to be.



Exercise 4.27. Let S be the set consisting of your group members. Find $f[S]$, where f is defined as in Example 4.24.

Exercise 4.28. Let $A = \{a,b,c,d,e\}$ and $B = \{1,2,3,4,5,6,7\}$. Also, suppose the function f is given by the subset of $A \times B$, $f = \{(a,5),(b,7),(c,4),(d,1),(e,4)\}$. Let S be the set $\{a,c,e\}$. Find $f[S]$.

Exercise 4.29. Consider the function f from Exercise 4.28. Let $S = \{a,b,c\}$ and $T = \{b,d\}$. Compute:

1. $f[S]$
2. $f[T]$
3. $S \cup T$
4. $f[S \cup T]$
5. $f[S] \cup f[T]$
6. $S \cap T$
7. $f[S \cap T]$
8. $f[S] \cap f[T]$
9. $S - T$
10. $f[S - T]$
11. $f[S] - f[T]$

Compare part 4 with part 5, part 7 with part 8, and part 10 with part 11.

Problem 4.30. *Let f be as in Example 4.28, and let $S = \{a,b,c\}$ and $T = \{b,e\}$.*

1. *Is $f[S\cup T] = f[S]\cup f[T]$? Is one a proper subset of the other?*
2. *Is $f[S\cap T] = f[S]\cap f[T]$? Is one a proper subset of the other?*
3. *Is $f[S-T] = f[S]-f[T]$? Is one a proper subset of the other?*

Problem 4.31. [33]

Give examples of the following (remember that as always, you must include justification with correct terminology):

1. *A function f, and nonempty subsets S and T of the domain of f, such that $f[S\cap T] = \emptyset = f[S]\cap f[T]$.*
2. *A function f, and nonempty subsets S and T of the domain of f, such that $f[S\cap T] = \emptyset$, but $f[S]\cap f[T] \neq \emptyset$.*
3. *A function f, and nonempty subsets S and T of the domain of f, such that $f[S]-f[T] = \emptyset$, but $f[S-T] \neq \emptyset$.*
4. *A function f and nonempty subsets S and T of the domain of f, such that $|f[S]| = |f[T]|$ but $f[S] \neq f[T]$.*

It is now time to start proving/disproving the statements we've been examining in the last few problems. For the next three problems, prove the true one(s) and disprove the false one(s). For the false one(s), create your own counterexamples from scratch, as opposed to using ones from previous problems (it's good practice).

Problem 4.32. [34]

Make sure you have read the paragraph directly above this before starting.

1. $f[S\cap T] \subseteq f[S]\cap f[T]$
2. $f[S]\cap f[T] \subseteq f[S\cap T]$

Problem 4.33.

1. $f[S-T] \subseteq f[S]-f[T]$
2. $f[S]-f[T] \subseteq f[S-T]$

Problem 4.34.

1. $f[S\cup T] \subseteq f[S]\cup f[T]$
2. $f[S]\cup f[T] \subseteq f[S\cup T]$

Definition 4.35. *Let A and B be sets. Let $f : A \to B$ be a function, and let $T \subseteq B$. Then $f^{-1}[T]$ is defined to be the set of pre-images of elements of the set T, i.e.*

$$f^{-1}[T] = \{x \in A : f(x) = y \text{ for some } y \in T\}.$$

Be careful: $f^{-1}[T]$ is a set.

Exercise 4.36. Let $A = \{a,b,c,d\}$, $B = \{1,2,3,4\}$, and let $f : A \to B$ be defined by $\{(a,3),(b,1),(c,3),(d,2)\}$. Find $f^{-1}[\{2\}]$, $f^{-1}[\{3\}]$, $f^{-1}[\{2,3\}]$, $f^{-1}[\{4\}]$, and $f^{-1}[\{3,4\}]$.

Problem 4.37.

1. *Let A and B be sets. Let f be a function from A to B, and let S be a finite subset of the range of f. Decide and justify using correct terminology which of the following is the correct statement (without knowing any more information about S or f):*
   (a) $|f^{-1}[S]| = |S|$
   (b) $|f^{-1}[S]| < |S|$
   (c) $|f^{-1}[S]| > |S|$
   (d) $|f^{-1}[S]| \geq |S|$
   (e) $|f^{-1}[S]| \leq |S|$
   (f) *No relationship can be made between $|f^{-1}[S]|$ and $|S|$*
2. *Now suppose that S is a finite subset of B. (How is this different from part 1?) Which is the correct statement now? (Same as before or not?)*
   (a) $|f^{-1}[S]| = |S|$
   (b) $|f^{-1}[S]| < |S|$
   (c) $|f^{-1}[S]| > |S|$
   (d) $|f^{-1}[S]| \geq |S|$
   (e) $|f^{-1}[S]| \leq |S|$
   (f) *No relationship can be made between $|f^{-1}[S]|$ and $|S|$*

4.3 One-to-one and Onto Functions

Definition 4.38. *A function f is one-to-one (1-1), or injective, if the following holds: $\forall x_1 \forall x_2 \, [f(x_1) = f(x_2) \to x_1 = x_2]$ where x_1 and x_2 are elements of the domain of f.*

Exercise 4.39. Translate the definition of "one-to-one" into words, and interpret. Make sure you understand and haven't just translated literally and stopped.

Exercise 4.40. Let $A = \{1,2,3\}$ and $B = \{\text{dog, cat, mouse}\}$, and consider the function f given by the set $\{(1,\text{dog}),(2,\text{cat}),(3,\text{cat})\}$. Check the definition of one-to-one carefully on this function, to see if it is one-to-one or not.

Exercise 4.41. Let $A = \{1,2,3\}$ and $B = \{a,b,c,d\}$. Find both a one-to-one function and a non one-to-one function using this domain and codomain (use ordered pair notation).

Exercise 4.42. Let $A = \{a,b,c,d\}$ and $B = \{1,2,3\}$. Is it possible to have a one-to-one function with this domain and codomain? Explain.

Exercise 4.43. (PD) Give an interval A on which the following functions are one-to-one:

1. $F : A \to \mathbb{R}$, $f(x) = \cos x$
2. $F : A \to \mathbb{R}$, $f(x) = \cos(2x)$
3. $F : A \to \mathbb{R}$, $f(x) = x^3 - 3x^2 + x + 1$

Definition 4.44. *Let A and B be sets. A function $f : A \to B$ is called onto or surjective if for every element $b \in B$ there is an element $a \in A$ such that $f(a) = b$. A function f is called a surjection if it is onto.*

Exercise 4.45. Translate the definition of "onto" into symbols.

Exercise 4.46. Let the set $A = \{1,2,3\}$ and $B = \{a,b,c\}$. Is the function given by the set $\{(1,a),(2,b),(3,b)\}$ onto? Why or why not? If not, find a function from A to B that is onto.

Exercise 4.47. Let $A = \{1,2,3\}$ and $B = \{a,b,c,d\}$. Is it possible to have an onto function with this domain and codomain? Explain.

Observe: A function is ALWAYS onto its range by definition.

Exercise 4.48. (PD) Give an interval B such that the following functions are onto:

1. $F : \mathbb{R} \to B$, $f(x) = \cos x$
2. $F : \mathbb{R} \to B$, $f(x) = \cos(2x)$
3. $F : \mathbb{R} \to B$, $f(x) = x^3 - 3x^2 + x + 1$

Exercise 4.49. How do one-to-one and onto relate to existence and uniqueness proofs? Why are these concepts in the same chapter?

Problem 4.50. [35]
Let $A = \{a,b,c\}$, and consider the function $C : P(A) \to \mathbb{Z}$ given by $C(S) = |S|$.

1. *Write out the domain of C (in roster notation).*
2. *What is the range of C?*
3. *Is C one-to-one?*
4. *Is C onto?*

Exercise 4.51. Look back at Problem 4.6, in particular part 2. How is that related to "onto"?

Problem 4.52. *Consider the function $f : 2\mathbb{Z} \to \mathbb{Z}$ given by $f(x) = \frac{x-10}{2}$, where $2\mathbb{Z}$ designates the set of even integers.*

1. *Prove that f is one-to-one.*
2. *Prove that f is onto. (Hint: consider Problem 4.6.)*

Problem 4.53. *Determine if $f : \mathbb{Z}^+ \to \mathbb{Z}^+$ given by $f(x) = |x+1|$ is one-to-one and/or onto, and prove your claims.*

Problem 4.54. [36] *Determine if $f : \mathbb{N} \to \mathbb{N}$ given by $f(x) = |x-1|$ is one-to-one and/or onto, and prove your claims.*

Problem 4.55. [37] *Let $\mathscr{F}_0$ be the set of functions $f : \mathbb{R} \to \mathbb{R}$ whose derivative at 0 exists. Let $G : \mathscr{F}_0 \to \mathbb{R}$ be given by $G(f) = f'(0)$. (So G maps each function f to its derivative at 0.)*

1. *Compute $G(3e^x + x^2)$ and $G(\ln(x+1)+7)$.*
2. *Determine if G one-to-one and/or onto, and prove all claims.*

Definition 4.56. *The function f is a <u>bijection</u> if it is both one-to-one and onto (both an injection and a surjection).*

Problem 4.57. *Prove that every nonconstant linear function is a bijection from $\mathbb{R} \to \mathbb{R}$.*

Problem 4.58. *Prove or disprove the following (note that $\mathbb{R}^2$ means $\mathbb{R} \times \mathbb{R}$):*

1. *The function $f : \mathbb{R}^2 \to \mathbb{R}$ given by $f(x,y) = x - 2y$ is a bijection.*
2. *The function $f : \mathbb{R} \to \mathbb{R}^2$ given by $f(x) = (x+1, x-1)$ is a bijection.*

Problem 4.59. *Prove or disprove: The function $f : \mathbb{R}^2 \to \mathbb{R}^2$ given by $f(x,y) = (x-y, 3x)$ is a bijection.*

4.4 Inverse Functions and Composition of Functions

4.4.1 Inverse Functions

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Definition 4.60. *Let f be a bijection from the set A to the set B. The inverse function of f is the function that assigns to an element $b \in B$ the unique element $a \in A$ such that $f(a) = b$. The inverse function of f is denoted f^{-1}, and we thus write*

$$f^{-1}(b) = a \leftrightarrow f(a) = b.$$

If f^{-1} exists, we say f is invertible.

Exercise 4.61. Why does it make sense that f must be a bijection in Definition 4.60?

Remark 4.62. Note from Definition 4.60 that the only requirement for f^{-1} to exist is for f to be a bijection, and if f^{-1} exists, f MUST be a bijection. So saying that f is a bijection is actually equivalent to saying it is invertible.

Exercise 4.63. How is Definition 4.60 different from Definition 4.35?

Remark 4.64. You are no doubt already familiar with inverse functions from precalculus. For example, $f(x) = mx + b$ and $g(x) = \frac{x-b}{m}$ (if $m \neq 0$)) are inverses of each other, as are $f(x) = e^x$ and $g(x) = \ln x$, $f(x) = x^3$ and $g(x) = \sqrt[3]{x}$, etc.

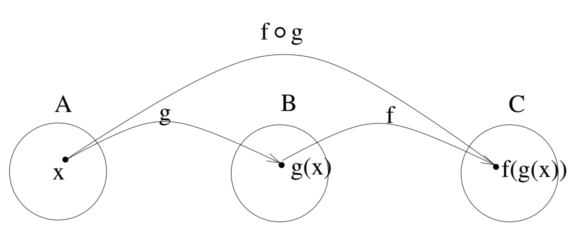
Problem 4.65. *Determine which functions are invertible, and for the invertible ones find their inverse functions:*

1. $f : \mathbb{R}^2 \to \mathbb{R}$ *given by* $f(x,y) = xy^2$
2. $g : \mathbb{Z}^2 \to \mathbb{Z}^2$ *given by* $g(x,y) = (y+1, 2x)$ *($\mathbb{Z}^2$ is $\mathbb{Z} \times \mathbb{Z}$)*
3. $h : \mathbb{R}^2 \to \mathbb{R}^2$ *given by* $h(x,y) = (y+1, 2x)$

4.4.2 Composition of Functions

Definition 4.66. *Let A, B, and C be sets. Let $g : A \to B$ be a function and let $f : B \to C$ be a function. The composition of the functions f and g, $f \circ g : A \to C$, is defined by $(f \circ g)(a) = f(g(a))$ for every $a \in A$.*

We can picture composition as follows:



Exercise 4.67. Let $A = \{1,2,3\}$, $B = \{a,b,c,d\}$, and $C = \{\text{dog, cat, mouse, bear}\}$. Let $f : B \to C$ be given by $\{(a,\text{cat}),(b,\text{bear}),(c,\text{mouse}),(d,\text{cat})\}$ and $g : A \to B$ be given by $\{(1,b),(2,d),(3,a)\}$. Compute $f \circ g$, writing your answer in ordered pair notation.

Problem 4.68. [39]

1. *Construct sets A,B, and C, and functions f and g that show that $f \circ g$ can be one-to-one even if f isn't.*

2. *Construct sets A,B, and C, and functions f and g that show that $f \circ g$ can be onto even if g isn't.*

Problem 4.69. *Let A, B, and C be sets, and let $f : B \to C$ and $g : A \to B$ be functions. Prove that if f and g are one-to-one then $f \circ g$ is one-to-one.*

Problem 4.70. *Let A, B, and C be sets, and let $f : B \to C$ and $g : A \to B$ be functions. Prove that if f and g are onto then $f \circ g$ is onto.*

Problem 4.71. *Prove that there exists a unique function $h : \mathbb{R} \to \mathbb{R}$ such that $(f \circ h)(x) = (h \circ f)(x) = f(x)$ for all $f : \mathbb{R} \to \mathbb{R}$.*

Chapter 5

Induction

5.1 Introduction to Induction

Mathematical induction is a technique of proof that we use when we want to prove a statement holds for every positive integer or every member of a countable (listable) set in general (as opposed to the set of real numbers for example, which we can't write in a list) [1]. We often think of this as a generalization of showing no matter how long a stack of dominoes is, all will eventually be knocked down as long as the first one is. We do this using the following procedure:

(1) Show that the first domino can be knocked down.
(2) Show that if we know that any domino anywhere in the stack will eventually be knocked down, known as the kth domino, the next one will also be knocked down, which is the $(k+1)$st domino.

Notice that since the first one can be knocked down by (1), (2) tells us that the second one will also be knocked down. Now that we know that the second one will be knocked down, (2) tells us that the third one will be knocked down. Once we know that the third one can be knocked down, (2) tells us that the fourth one will be knocked down, and so on, which we interpret as saying that all dominoes will eventually be knocked down for a stack of any (finite) length. Let's work through an exercise to get you thinking this way.

Exercise 5.1. 1. Suppose you have two math textbooks and you place them in a stack on your desk. How many possible orders are there for the way they are placed?

2. What if you have three textbooks? How many orders then? Four textbooks?

3. Can you use what you already knew for two textbooks to more easily find the amount of orders for three, without having to count from scratch? And then

[1] We say that the set of real numbers is *uncountable*, meaning there does not exist a bijection between $\mathbb{N}$ and $\mathbb{R}$. There is a nice presentation of the standard argument we use to prove this fact here.

use that amount to more easily find the amount for four (again without having to count from scratch)? If you knew how many orders there were for say, 100 books, could you then use that to determine how many orders there were for 101?

4. Let's generalize that last question: If you knew how many orders there were for k textbooks, could you use that to determine how many orders there are for $k+1$? (This is an important part of the induction process, so don't gloss over it.)

5. State the result you have concluded from this exercise, as in, "The number of possible orders for n textbooks to be placed in a stack is ____."

We now generalize the process.

Definition 5.2. *Induction*
Suppose we have a mathematical statement $P(n)$ that we would like to show is true for every positive integer n (this is the equivalent of saying that the nth domino will be knocked down for any n). We perform the following steps:

(1) Show P(1) is true (Base case or Basis step)
(2) Show that $P(k) \to P(k+1)$ for $k \geq 1$ (Inductive Step; $P(k)$ is called the induction hypothesis)

Remark 5.3. Note that (2) is a conditional statement, so we prove it as we would any conditional statement. Note also that in the base case, we usually start with $P(1)$, but sometimes we need a different base case, or sometimes we even need more than one.

Exercise 5.4. Consider your statement from Exercise 5.1, part 5, as $P(n)$. (Recall that $P(n)$ is an entire proposition, so it requires a subject and verb, not just a subject or object.)

1. What is the statement $P(1)$?

2. Write out the inductive statement $P(k) \to P(k+1)$ using the $P(n)$ you just found.

Remark 5.5. If you have a computer science background, you are no doubt noticing the similarity to recursion. They are essentially the same overall concept in our context. But notice that one big difference here is that we are counting up, not down.

We now look at another example together. Note that we will carefully identify the statement $P(n)$, and it is a proposition with a subject and predicate, not just a quantity.

Example 5.6. [40] In this example, we will prove by induction that $n < 2^n$ for all $n \geq 1$. *Everything in italics below is side commentary, and would not be included in our final write-up.*
Proof: Let $P(n)$ be the statement $n < 2^n$.

We start with our base case, which should be fairly straightforward:

Base Case: We show that $P(1)$ holds:

$2^1 = 2$, and since $1 < 2$, $P(1)$ holds.

We now move on to the inductive step, which we recall is the conditional statement $P(k) \rightarrow P(k+1)$. We recall also that $P(k)$ is our induction hypothesis.

Inductive Step: We assume the statement $P(k)$, which is the statement $k < 2^k$, holds for some $k \geq 1$. We will show that $P(k+1)$ holds, i.e. that $k+1 < 2^{k+1}$.

Here is where we need to be a bit clever. Let's think about where we have started, and where we want to end up. What we know is that $k < 2^k$, and what we want is that $k+1 < 2^{k+1}$. So how can we get from k to $k+1$, and from 2^k to 2^{k+1}?

We aren't going to necessarily be able to do that in one step, so let's just focus on one side, say the left. Getting from k to $k+1$ is easy - we just have to add 1. So, let's add 1 to both sides of the inequality $k < 2^k$ and see what we get.

Since $k < 2^k$ by the induction hypothesis, $k+1 < 2^k+1$ which we obtain by adding 1 to both sides.

Note $2^k + 1 \neq 2^{k+1}$ - if it did, we'd be done!

We have

$k+1 < 2^k+1$

So now the left side of the inequality is what we want, but the right side needs more work. This is the part of the proof that will involve the most playing around, experimentation, and scrap work. But, it might help to work backward from what we want, which is 2^{k+1}.

We know that $2^{k+1} = 2(2^k) = 2^k + 2^k$, so now getting from $2^k + 1$ to 2^{k+1} boils down to getting from $2^k + 1$ to $2^k + 2^k$. Certainly, these last two quantities are not *equal to each other. But, they don't have to be, because what we ultimately want is an inequality, not an equality.*

So how do 2^k+1 *and* 2^k+2^k *relate to each other? Since they're not equal, let's think about which one of them must be larger. It depends on what* k *is of course, but since we assume that* $k \geq 1$*, we know that* $2^k \geq 2$*. Thus, we can conclude that* $2^k+1 < 2^k+2^k$*.*

Since we had that $k+1 < 2^k+1$*, and we now know that* $2^k+1 < 2^k+2^k$*, we can combine those results to obtain* $k+1 < 2^k+2^k$*. Now, let's see how we can write this up cleanly and formally in a way that leads us to our final desired result.*

We have

$$\begin{aligned} k+1 &< 2^k+1 \\ &< 2^k+2^k \text{ since } k \geq 1 \text{ by assumption} \\ &= 2(2^k) \\ &= 2^{k+1}. \end{aligned}$$

Combining our inequalities, we obtain that $k+1 < 2^{k+1}$, which is what we needed to show. We have thus proved by induction that $n < 2^n$ for all $n \geq 1$.
■

Exercise 5.7. Observe that in the previous example, in our inductive step we decided to get from $k < 2^k$ to $k+1 < 2^{k+1}$ by focusing on the left side of the inequality and adding 1 to get from k to $k+1$. But, we could have just as easily focused on the right side of the inequality, aiming to get from 2^k to 2^{k+1}. Recalling that $2^{k+1} = 2(2^k)$, we see that we can get from 2^k to 2^{k+1} by multiplying by 2. So, try to redo the inductive step from Example 5.6 by instead multiplying both sides of $k < 2^k$ by 2 and see if you can figure out how to get to $k+1 < 2^{k+1}$ from there.

This is just one of the many, many different types of statements that can be proved by induction. Note that most of the work is in the inductive step, when you are trying to prove the statement $P(k+1)$. The algebra and/or reasoning can look very different from problem to problem, so you can only model the *structure* of your proofs off of examples, not the wording and not the individual algebraic steps. Each proof requires different decisions to reach the end goal.

Problem 5.8. *Prove by induction that* $2^n < n!$ *for all* $n \geq 4$*.*

Problem 5.9. *Prove by induction that the quantity* $3^n - 1$ *is even for all* $n \in \mathbb{Z}^+$*.*

Example 5.10. Consider the sequence defined by $a_0 = 0$, and $a_n = 5a_{n-1} + 1$ for all $n \geq 1$. This is known as a recursive sequence, where each term is defined by the previous one. We will see that this sequence can also be obtained directly from the formula $f(n) = \dfrac{5^n - 1}{4}$.

First let's understand what it is we're trying to prove.

Here we examine the recursive definition:

$a_0 = 0$

$a_1 = 5a_0 + 1 = 5(0) + 1 = 1$

$a_2 = 5a_1 + 1 = 5(1) + 1 = 6$

$a_3 = 5a_2 + 1 = 5(6) + 1 = 31$

$a_4 = 5a_3 + 1 = 5(31) + 1 = 156$

and so on ...

Here we compare the values produced by the formula:

$$f(0) = \frac{5^0 - 1}{4} = \frac{1-1}{4} = \frac{0}{4} = 0$$
$$f(1) = \frac{5^1 - 1}{4} = \frac{5-1}{4} = \frac{4}{4} = 1$$
$$f(2) = \frac{5^2 - 1}{4} = \frac{25-1}{4} = \frac{24}{4} = 6$$
$$f(3) = \frac{5^3 - 1}{4} = \frac{125-1}{4} = \frac{124}{4} = 31$$
$$f(4) = \frac{5^4 - 1}{4} = \frac{625-1}{4} = \frac{624}{4} = 156$$

We can see these all match the values from the recursive formulation on the left.

Hopefully this is somewhat convincing that our formula should hold for all n. But, we cannot just assume that; we have to *prove* it. This is where induction comes in, which you will do in the following problem.

Problem 5.11. *Let a_n be the sequence defined in Example 5.10. Recall that we observed that this sequence can seemingly be represented by the formula $f(n) = \frac{5^n-1}{4}$. Let $P(n)$ be the statement $a_n = f(n)$, for $n \in \mathbb{N}$, and prove this statement using induction.*

Problem 5.12. *Let a_n be the sequence defined recursively by $a_0 = 1$, $a_{n+1} = 3 - \frac{1}{a_n}$. Prove that $a_n \geq 2$ for all a_n with $n \geq 1$.*

5.1.1 Summations

Definition 5.13. *Recall from Calculus I that*

$$\sum_{i=m}^{n} a_i = a_m + a_{m+1} + a_{m+2} + \cdots + a_n, \text{ where } i \text{ is called the } \underline{\text{index}}.$$

This is called "Σ"-notation ("sigma notation") or summation notation. We can think of the index "i" as the "counter", and we can of course use any letter we would like for the index. The letters i, j, and k are all common choices. Note that the first term comes from the number m below the sigma and the last term comes from the number n above the sigma. Usually m is 1 or 0.

Remark 5.14. If you have taken Calc II, you will probably recall that the sums were usually infinite, but they will not be in our setting.

Example 5.15. Consider the following sums:

$$\sum_{i=1}^{n} i = 1+2+3+\cdots+n$$
$$\sum_{i=1}^{n} i^2 = 1+4+9+\cdots+n^2$$
$$\sum_{i=1}^{6} i^2 = 1+4+9+16+25+36 = 91$$
$$\sum_{i=2}^{6} i^2 = 4+9+16+25+36 = 90 = \left(\sum_{i=1}^{6} i^2\right) - 1 = \sum_{j=1}^{5} (j+1)^2 \text{ (change of index, } j = i-1)$$
$$\sum_{i=1}^{6} 4 = 4+4+4+4+4+4 = 6(4) = 24$$

Exercise 5.16. Compute the following sums:
$\sum_{i=1}^{10} i$, $\sum_{i=1}^{10} i^2$, $\sum_{i=1}^{5} 2i^3$, $\sum_{i=4}^{10} \frac{1}{i}$

What if we want to compute $\sum_{i=1}^{100} i$? This is much more computation than we care to do! So instead, we are lucky enough to have the following formulas:

Theorem 5.17. *The following formulas hold for all $n \in \mathbb{Z}^+$:*

$$\sum_{i=1}^{n} i = 1+2+3+\cdots+n = \frac{n(n+1)}{2}$$
$$\sum_{i=1}^{n} i^2 = 1^2+2^2+3^2+\cdots+n^2 = \frac{n(n+1)(2n+1)}{6}$$
$$\sum_{i=1}^{n} i^3 = 1^3+2^3+3^3+\cdots+n^3 = \frac{n^2(n+1)^2}{4}$$

Proof: Left as problems.

Exercise 5.18. Verify that each of the formulas in Theorem 5.17 works for several values of n.

You will prove that two of these formulas hold for all n as problems below (the other one can be verified in the same way). [41]

Problem 5.19. *Prove that* $\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$ *for all* $n \in \mathbb{Z}^+$.

Problem 5.20. *Prove that* $\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$ *for all* $n \in \mathbb{Z}^+$.

Problem 5.21. *Prove that* $\sum_{i=0}^{n} \frac{1}{3^i} = \frac{3^{n+1}-1}{2(3^n)}$ *for all* $n \in \mathbb{N}$.

Problem 5.22. *Use induction to prove that if* $a, r \in \mathbb{R}$ *and* $r \neq 0$*, then*

$\sum_{i=0}^{n} ar^i = a\left(\frac{r^{n+1}-1}{r-1}\right)$ *if* $r \neq 1$. *If you have taken Calc II you will recognize this as a geometric series (the finite version).*

Problem 5.23. *Prove that if* $n \in \mathbb{Z}^+$*, then* $2 \cdot 6 \cdot 10 \cdot 14 \cdot \cdots \cdot (4n-2) = \frac{(2n)!}{n!}$. *(There is a similar notation for products as for sums using* Π *if you would like to use it.)*

Problem 5.24. *Prove that* $\left(1-\frac{1}{4}\right)\left(1-\frac{1}{9}\right)\cdots\left(1-\frac{1}{n^2}\right) = \frac{n+1}{2n}$. *(Hint: you have to figure out what your base case is here.)*

Chapter 6

Division and Modular Arithmetic

6.1 The Division Algorithm

Definition 6.1. *If a and b are integers, we say that <u>a divides b</u> if there exists an integer c such that $b = ac$. The notation $a|b$ denotes that a divides b, and we write $a \nmid b$ when a does not divide b.*

Remark 6.2. When a divides b we may also say that a is a *factor* of b and that b is a *multiple* of a, and that b is *divisible by a*. BUT, the main language we should defer to is the language given in Definition 6.1.

Remark 6.3. CAREFUL: $a|b$ is a *proposition* (because it has a verb in it, "divides"); $\frac{b}{a}$ is a noun (a number), and so they are not equivalent. $a|b$ has a truth value, $\frac{b}{a}$ does not. At no point in this chapter should you need to invoke $\frac{b}{a}$. Also, since $a|b$ is a proposition, you should never be writing that it is "=" to anything, since propositions aren't *equal*.

Exercise 6.4. Does $5|20$? Does $(-5)|20$? Does $5|(-20)$? Does $20|5$? Does $0|0$? Justify all of these by carefully applying Definition 6.1.

Exercise 6.5. Show that for all $n \in \mathbb{Z}$, $n|0$. Does $0|n$?

Problem 6.6. [42] *Let a and b be integers. Prove that if $a|b$, then $a|bm$ for any integer m.*

Problem 6.7.

1. *Prove or disprove: for integers a, b, and c, if $a|b$ and $a|c$ then $a|(b+c)$.*
2. *Prove or disprove: for integers a, b, and c, if $a|(b+c)$, then $a|b$ and $a|c$.*

Problem 6.8.

1. *Prove or disprove: for integers a, b, and c, if $a|c$ and $b|c$ then $a|b$.*
2. *Prove or disprove: for integers a, b, and c, if $a|b$ and $b|c$ then $a|c$.*

Problem 6.9. *Use <u>results</u> from above (<u>not</u> Definition 6.1) to prove the following two statements:*

1. *If x, y, z are integers such that $x|y$ and $x|z$, then $x|(sy+tz)$ for any integers s and t.*
2. *For integers n, x, y, z, if $n|x$ and $n|z$, and $x+y=z$, then $n|y$.*

Problem 6.10. *Use induction to prove that $3|(n^3+2n)$ for all $n \in \mathbb{Z}^+$.*

Problem 6.11. [43] *Let n be a two-digit positive integer. Prove that $3|n$ if and only if 3 divides the sum of n's digits. (You probably know that this is true for all positive integers, not just two-digit ones, but for now let's just prove it for two digits.)*

So far, we have studied situations where one integer is a multiple of another, meaning if we divide one by the other we have no remainder. We now consider when the remainder may be nonzero when we divide one integer by another, for example if we divide 25 by 7.

Theorem 6.12. *The Division Algorithm*
Let a be an integer and d be a positive integer. Then there are unique integers q and r with $0 \leq r < d$ such that $a = dq + r$.
We will not prove this here.

Definition 6.13. *In the division algorithm equation, d is the divisor, a is the dividend, q is the quotient, and r is the remainder.*

Exercise 6.14. In each of the following, divide a by d by writing the appropriate equation according to the division algorithm (making sure your r and q conform to the requirements):

1. $a = 13, d = 5$
2. $a = 15, d = 5$
3. $a = 3, d = 5$
4. $a = -3, d = 5$

As before, note that we are actually expressing division as multiplication. Hopefully you can see that even though you may not be used to writing it this way, this version of "division" does conform to your previous understanding. Moving forward, when we say we "divide" one integer a by another integer d, we will always mean according to the equation from division algorithm.

Exercise 6.15. Note that in the division algorithm, if we did not have the condition $0 \leq r < d$, we could not say that the integers q and r are unique. Why?

Exercise 6.16. Recall that in Exercise 2.21, we showed that an integer must be even or odd, and in Problem 2.41, we showed that it cannot be both. The Division Algorithm actually establishes both of these results simultaneously. How/why?

Exercise 6.17. Given an arbitrary integer a, the integer 3 may or may not divide it. What are the possible remainders when a is the dividend and 3 is the divisor? What are the possible corresponding division algorithm equations? How many are there?

Problem 6.18. *(DC)*

1. *Prove that for any set of three consecutive integers, 3 must divide at least one of them.*
2. *Prove that* $3|(n^3+2n)$ *for all* $n \in \mathbb{Z}^+$ *without using induction, by using the fact that* $n^3+2n = n(n+1)(n+2) - 3n^2$.

Problem 6.19. *Prove that for all* $n \in \mathbb{Z}$, $3 \nmid n^2+1$. *(Hint: use a proof by cases on* n *combined with a proof by contradiction.)*

Definition 6.20. *The quantity* $r = a \bmod d$ *is the unique remainder* r *when* a *is divided by* d *according to the division algorithm.*

Exercise 6.21. Evaluate the following by using the division algorithm equation:

1. 23 mod 10
2. 10 mod 23
3. -23 mod 10

Problem 6.22. *Let* d, a *be integers with* $d > 0$. *Prove that* $d|a$ *if and only if* $a \bmod d = 0$.

6.2 Modular Arithmetic

We will now consider the situation when two integers are not necessarily multiples of the same number m, but their distance from each other is still a multiple of m. For example, observe that 27 and 43 are not multiples of 4; however, the separation between them $(27-43=-16)$ is a multiple of 4. This will tell us that these numbers are "related" in a way that will turn out to be important in the study of number theory.

Definition 6.23. *If* a *and* b *are integers and* m *is a positive integer, then* <u>*a is congruent to b modulo m*</u> *if* $m|a-b$. *We use the notation* $a \equiv b \pmod{m}$ *and usually say "a is congruent to b mod m". If a and b are not congruent modulo (mod) m, we write* $a \not\equiv b \pmod{m}$.

Remark 6.24. Careful: Definition 6.23 is different from Definition 6.20. a mod m is a quantity, whereas $a \equiv b \pmod{m}$ is a proposition. The two are certainly related, which we will show in Theorem 6.34. But before we get there, it is important to understand each one separately. You can think of $a \equiv b \pmod{m}$ as saying a is congruent to b, and then the (mod m) is just a parenthetical remark specifying the way in which they are congruent. The quantity a mod m is just a number.

Exercise 6.25. Use Definition 6.23 to determine if the following hold:

1. $51 \stackrel{?}{\equiv} 8 \pmod{11}$
2. $51 \stackrel{?}{\equiv} -8 \pmod{11}$
3. $151 \stackrel{?}{\equiv} 19 \pmod{11}$
4. $1005 \stackrel{?}{\equiv} 361 \pmod{14}$

Exercise 6.26. Find a $b \in \mathbb{Z}$ such that $b \equiv 3 \pmod 5$ and $b \equiv 2 \pmod 7$. Is this b unique?

Problem 6.27. *Let m be a positive integer.*

1. *Show that if $a \equiv b \pmod m$, then $a+k \equiv b+k \pmod m$ for any $k \in \mathbb{Z}$.*
2. *Show that if $a \equiv b \pmod m$ and $c \equiv d \pmod m$ for $a,b,c,d \in \mathbb{Z}$, then $a+c \equiv b+d \pmod m$.*

Problem 6.28. *Let m be a positive integer. Prove that if $a \equiv b \pmod m$ and $c \equiv d \pmod m$ for $a,b,c,d \in \mathbb{Z}$, then $ac \equiv bd \pmod m$.*

Problem 6.29. *Prove that the relationship of being congruent mod m for $m \in \mathbb{Z}^+$ is an "equivalence relation" on the integers; that is, a relationship between integers that satisfies the following three properties:*

1. *$a \equiv a \pmod m$ for all $a \in \mathbb{Z}$ (reflexive property)*
2. *If $a \equiv b \pmod m$, then $b \equiv a \pmod m$, for all $a,b \in \mathbb{Z}$ (symmetric property)*
3. *If $a \equiv b \pmod m$ and $b \equiv c \pmod m$, then $a \equiv c \pmod m$ (transitive property)*

Problem 6.30. *Let m be a positive integer and let $a,b,n \in \mathbb{Z}$. Prove or disprove the following:*

1. *If $a \equiv b \pmod m$ then $an \equiv bn \pmod m$.*
2. *If $an \equiv bn \pmod m$ then $a \equiv b \pmod m$.*

Problem 6.31. [44] *Prove that for all $n \in \mathbb{Z}$, $3 \nmid n^2+1$ using the language of modular arithmetic and results from Problems 6.27 - 6.30 (so using different notation then you would have in Problem 6.19). It is once again suggested to use a proof by cases on n.*

Though Definition 6.23 is the official definition of being congruent modulo m, we have another way to interpret and apply this concept. The next lemma and problem, and the theorem that follows them, will lead us to this interpretation.

Lemma 6.32. *Let $m \in \mathbb{Z}^+$ and $d \in \mathbb{Z}$. If $|d| < m$ and $m|d$, then $d = 0$.*
Proof: By contradiction.

Suppose that $d \neq 0$. Since $m|d$, $d = km$ for some $k \in \mathbb{Z}$, by Definition 6.1, and $k \neq 0$ since $d \neq 0$. Then because $k \in \mathbb{Z}$, $|k| \geq 1$ so

$|d| = |km|$
$= |k||m|$
$\geq |m|$
$= m$ *since* $m \in \mathbb{Z}^+$

But this contradicts that $|d| < m$. $\rightarrow\leftarrow$ Thus, $d = 0$.
■

Problem 6.33. *Let $m, a \in \mathbb{Z}$ with $m > 0$. If $m|a$ and $a = mq + s$ for some integers q and s, then $m|s$.*

Theorem 6.34. *Let a and b be integers and let m be a positive integer. Then $a \equiv b(mod\ m)$ if and only if a mod $m = b$ mod m.*

Proof: ($\leftarrow$) If a mod $m = b$ mod m, then $a \equiv b(mod\ m)$.
Let a, b, m be integers with $m > 0$, and suppose that a mod $m = b$ mod m. Then by definition of a mod m,

$a = mq + r$
and
$b = mp + r$

with p, s, r being integers and $0 \leq r < m$ (same r for both by supposition). Then

$a - mq = r$
and
$b - mp = r$ *so*
$a - mq = b - mp$

Rearranging and factoring gives $a - b = m(q - p)$. Since q and p are integers, $q - p$ is an integer by the Axiom of Closure for Integers, so $m|a - b$ by Definition 6.1. Thus $a \equiv b(mod\ m)$ by definition of congruence mod m.

($\rightarrow$) If $a \equiv b(mod\ m)$, then a mod $m = b$ mod m.
Let a, b, m be integers with $m > 0$, and suppose that $a \equiv b(mod\ m)$. Then $m|(a - b)$ by definition of congruence mod m. We can divide a and b by m according to the division algorithm to obtain

(1) $a = qm + r$

where $q, r \in \mathbb{Z}, 0 \leq r < m$, and

(2) $b = pm + s$

where $p, s \in \mathbb{Z}$, $0 \leq s < m$. Our goal is to show $r = s$.

Subtracting equation (2) from (1) yields

$a - b = qm + r - (pm + s)$, *or*
$a - b = (q - p)m + r - s$

Since $q, p, r, s \in \mathbb{Z}$, so are $q - p$ and $r - s$ by the Axiom of Closure for Integers. We know that $m | a - b$, so by Problem 6.33, $m | r - s$.

Since $0 \leq r < m$ and $0 \leq s < m$ and $r, s \in \mathbb{Z}$, the largest $|r - s|$ can be is if $r = m - 1$ and $s = 0$ or vice versa, making $|r - s|$ be at most $m - 1$. Thus, $|r - s| < m$.

But we already concluded that $m | r - s$, so by Lemma 6.32, $r - s = 0$. Thus $r = s$, and since $r = a \bmod m$ and $s = b \bmod m$, $a \bmod m = b \bmod m$.
■

Exercise 6.35. Use Theorem 6.34 to verify your answers to Exercise 6.25:

1. $51 \stackrel{?}{\equiv} 8 \pmod{11}$

2. $51 \stackrel{?}{\equiv} -8 \pmod{11}$

3. $151 \stackrel{?}{\equiv} 19 \pmod{11}$

4. $1005 \stackrel{?}{\equiv} 361 \pmod{14}$

Problem 6.36. 1. *Prove by induction and previous results that if $a \bmod m = r$ for an integer a and positive integer m, then $a^n \bmod m = r^n \bmod m$ for $n \in \mathbb{Z}^+$.*

2. *Use the result of part (1) to compute the remainder when 24^{24} is divided by 7.*

Remark 6.37. If you were wondering why we think of modular congruence in two different ways, hopefully Problem 6.36 will give you an idea. Depending on the context, one interpretation might be easier or more useful to work with, and in particular there are many computational applications that arise from the result of Theorem 6.34.

6.3 Prime Numbers

We end our course with a brief discussion of prime numbers. In particular, we will prove one of the most fundamental theorems in mathematics, a result which has been known for over 2000 years!

Definition 6.38. *A positive integer p greater than 1 is called prime if the only positive factors of p are 1 and p. A positive integer that is greater than 1 and is not prime is called composite.*

We will need the following theorem to prove our main result.

Theorem 6.39. *The Fundamental Theorem of Arithmetic Every positive integer greater than 1 can be written uniquely as a prime or as the product of two or more primes where the prime factors are written in order of nondecreasing size.*
We will not prove this now.

Example 6.40. $132 = 2 \cdot 2 \cdot 3 \cdot 11 = 2^2 \cdot 3 \cdot 11$.

We are now ready to prove our final theorem of the course. The proof given below is from Euclid! The strategy is to assume that there are finitely many primes so we can list them all, but then show that that leads to a contradiction, so there must in fact be another one.

Theorem 6.41. *There are infinitely many primes.*
Proof: By contradiction.
Suppose there are only finitely many primes, $p_1, p_2, \ldots, p_n$.

Let $Q = p_1 p_2 \ldots p_n + 1$.

By the fundamental theorem of arithmetic, Q is prime or it can be written as the product of two or more primes. But Q cannot be prime because it is larger than any p_j so it is not equal to any p_j, and we assumed we had counted all the primes. So Q must be composite.

Thus, Q is a multiple of at least one prime, say p_j, i.e. $p_j | Q$. Since $Q = p_1 p_2 \ldots p_n + 1$, we have that

$$Q - p_1 p_2 \ldots p_n = 1$$

We know that p_j divides Q and by Problem 6.6 p_j divides $p_1 p_2 \ldots p_n$, so p_j divides $Q - p_1 p_2 \ldots p_n$ by Problem 6.9. Thus $p_j | 1$ which is not possible. $\rightarrow\leftarrow$
■

Remark 6.42. The above theorem has given way to lots of modern computational strategies to find the "next" prime (if we didn't know there were infinitely many, that would be an exercise in futility!). At the time of publication of this text, the largest known prime had 41,024,320 digits. The proof above is also a nice example of what we call a *non-constructive* existence proof. Up until now, we have only used *constructive* existence proofs, meaning, we prove an object exists by producing that object. In the proof above, we prove that there must exist another prime other than the ones we listed (because if not, we have a contradiction). However, we didn't actually construct that prime - yet we still managed to prove that it has to exist! So, it is in fact possible to prove that an object exists without every actually producing it.

Chapter 7

Epilogue

Congratulations! You have made it through your transitions to advanced mathematics course. Let's revisit our quotation from the beginning:

Good mathematics is not about how many answers you know, but about how you behave when you don't know.

Hopefully you have a much better idea now about the "how to behave when you don't know"! That is, I hope you have learned a lot about how we work as mathematicians: how we carefully define our concepts, how we problem-solve, how we "check" ourselves to justify our own thinking and convince others that our conclusions are correct, and how we often have to grapple with concepts or problems for a long time before we understand them!

You can feel proud of all you have accomplished, and you are ready to move on to higher-level mathematics. Enjoy!

Notes to the Instructor

[1]This problem always takes a fair amount of discussion among the students, in particular the F/T case of course. I think it's important to let that play out organically and not "force" the right answer too quickly.

[2]I tend to have a lot of computer science majors in my class, and I find they can understand this material much better if they can understand it in the terms they are already familiar with from their computer science background.

[3]There is often disagreement on the truth of the hypothesis of this one and the conclusion of the next one, which is great for demonstrating that if the conclusion is true or the hypothesis is false, the other half doesn't matter.

[4]This problem is a good precursor to the idea of a counterexample.

[5]I usually have this problem be a current political statement (for example about an upcoming election, or about the handling of COVID in 2020-2022), that I've updated every semester. Here it's been made into a time-neutral (and less controversial) version for this publication. I do feel it's important to emphasize the real-world implications of correct and incorrect argumentation, especially given the current climate. This new version was inspired by an actual conversation I had regarding the 2021 inaugural poem by Amanda Gorman. I also am aware that this problem does not have a unique interpretation (for example, a student recently argued that the last two statements could not be considered because we did not have information about whether or not the measure of a poem's "goodness" is binary). Some might argue that the problem should be fixed because of this; I take the opposite stance. I think it makes the problem much more interesting and worthwhile if there is more than one interpretation!

[6]The parallel structure between this and Problem 1.64 is deliberate, and interestingly, students seem to find this one harder.

[7]Note that part 4 of this problem is a statement that will need to be negated in a proof by contradiction in Chapter 2.

[8]What I had mind by this question is along the lines of "no one nearby has it", "everyone nearby doesn't have it", etc., but students have provided other correct answers that I didn't think of.

[9]When I have students perform this negation in real analysis, it usually takes them at least 20 minutes.

[10]This is definitely the part of the notes with the most exposition, but I find that it does pay off later, in that very little exposition is required for the proofs in subsequent chapters.

[11]During group work and presentations, I often hear students using the language of the steps outlined above as well as the ones that follow below (e.g. “first we have to name our players”).

[12]I do find that it’s helpful to provide at least one example of correct wording. Otherwise, they really have no idea what I even mean.

[13]I do correct this in homework!

[14]The preliminary steps are here because I have found a non-trivial number of students negate a universally quantified statement to another universally quantified statement (without really realizing this is what they’re doing), and then believe that a counterexample suffices as the contradiction.

[15]Note that this problem refers back to Problem 1.36.

[16]I used to open with this example, but I found that too many students then tried to model every proof by contradiction after this one!

[17]I really like that this proof feels quite simple to them at this point, after a chapter of much harder ones. It’s a nice reminder of how far they’ve come.

[18]I say that this problem is in here “just for fun”, but it has turned out to be a valuable lesson in what it means for a proof to be wrong. Many students identify steps that they wouldn’t do or don’t understand, or that just look “weird”, as the incorrect one. I would say it is up to the instructor to decide, for proof 2, to accept if the mistake is in step 3 or step 5 (or both). I accept both, since it’s hard enough for them to even remember that the constant of integration is a thing, and when they do, I want to be encouraging, considering they did still capture the essence of the mistake.

[19]The last one always leads to a discussion of how 0 is also complex (and how again, we need to rely on the *definition*, not an informal understanding).

[20]I privilege the symbolic notation here to emphasize that this definition is a conditional statement, to be proved as any other conditional statement.

[21]A surprising number of students struggle with this question, resubmitting it several times.

[22]I use the words “and” and “or” in the definitions that follow rather than the symbols $\wedge$ and $\vee$ because I notice that, with the emphasis on the conditional statements which I would like to keep, students’ proofs in this section can get too symbol-heavy.

[23]I love this problem because it isn’t actually hard at all once you unpack the symbols, so it’s a great one to emphasize going back to the definitions and making sure you understand them.

[24]There is always a significant portion of the class that misinterprets sets A and B, concluding that statement (1) is true and (2) is false. The best way to interpret those statements is via the conditional definition, which is why I like this problem. The emphasis on the conditional statement is all to lead up to proofs involving sets.

[25]I have rewritten this portion of the notes (from here through to Problem 3.49) so many times, to get across the idea that subset statements are just conditional statements like any other they’ve seen. This is not necessarily the final iteration, but it is by far the most successful of all the ones I’ve had.

[26]So many students try to answer this question just by “looking”, without writing out any elements

of the sets. This is a good reminder to them not to skip that.

[27]This problem is clearly pretty simple logically, but I still like it because it requires students to notice that a proof by contradiction is probably the best way to go, and to interpret what it means to assume that two sets are not disjoint. If they do both of those things, they are then rewarded by an immediate contradiction.

[28]This is clearly a very open-ended question, and while I have had some students get an easy correct solution by just letting all the sets be empty, or equal to each other, that doesn't happen often enough for me to want to restrict it. I also like the idea of demonstrating that there can be low-hanging fruit if you look for it.

[29]This is hard, and I don't necessarily expect them to get it. But it's in here for completeness and to provide another challenge problem.

[30]I find that having already solved this problem makes surjectivity much easier for them to grasp, because it's something they already did.

[31]Even though existence and uniqueness can obviously be shown in one step by direct solving here, I prefer that the students usually split their proofs into separate existence and uniqueness portions, to get the feel for the structure of this type of proof. Once in a while there is a student that is thrown by the fact that the statement to be proved seems *too* obvious, but I find that usually the ease of the algebra allows them to focus on the structure.

[32]This is another problem I've gotten tons of different solutions for over the years, including many I didn't expect. This one is a perfect example of, if you give students the space to think for themselves, don't decide in advance which hints you think they need, and accept that a longer proof is still a valuable proof, you will get some great results.

[33]This is another problem is fairly easy computationally, but is here to reinforce the vocabulary and notation.

[34]In semesters with tight schedules (such as when there have been a lot of snow days), I have left the next three problems out. I like including them for the reinforcement of subset proofs and the function and set notation/concepts.

[35]The questions themselves in this problem are pretty easy to answer. The work here is in interpreting the notation.

[36]Students often get through an entire proof that this function is onto (by defining $x = y - 1$), which is a great reminder that the pre-image must be contained in the domain.

[37]This is a fun one because students' immediate reaction is often "Ugh! What is this?", but once they read it a little more closely they realize it isn't actually that hard.

[38]I acknowledge that this section is in here mostly out of a feeling of obligation to have mentioned it.

[39]You would be surprised at this point how many students construct examples at this point where elements of the domain don't have an image at all.

[40]I have gone back and forth about whether or not to provide examples here. I ultimately decided

to keep them because really, I don't want to spend a lot of time on induction and in particular on the correct formatting of an induction proof. I want students to master the overall idea so we can move on. But, it is most definitely a deliberate choice not to provide examples of every "type". It is also a deliberate choice to do the summation ones later, because in my experience students get locked into the idea that every inductive step involves "adding something to both sides" when those come first.

[41]You may not want to include all of these; I often don't.

[42]I always find it really exciting and rewarding when we get to this point in the material, and the students find these proofs really easy. And, by this time a lot of the structure and wording kinks have been ironed out and the proofs themselves are pretty well written. It's a nice demonstration of how far they've come.

[43]I find it surprising that many students have never heard of this rule. And that many (even some of the CS majors) report that they have never thought of the digits of a number this way.

[44]This problem is a bit of a jump in level, but it can be a good "challenge" problem for those who are ahead.